Moon Formation by Triple Phase Transition in the Differentiating Proto-Earth

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Abstract. The origin of the Moon remains one of the unresolved problems of planetary science. The canonical giant-impact model faces growing geochemical difficulties: it does not naturally predict the near-identical Earth–Moon isotopic signature, the lunar crustal dichotomy, or the $\approx$ 350 Myr delay of the terrestrial dynamo. This work proposes an alternative grounded in a necessary thermodynamic observation: every Earth-mass planet emerges from accretion in a state of near-total silicate-mantle melting, accretion energy exceeding the total melting energy by a factor of $\approx$ 155 (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The proto-Earth is therefore a rapidly rotating magmatic body ($T_{\mathrm{rot}} \approx 3.5$ h — a value constrained directly by the terrestrial-planet accretion literature, not derived from a contraction argument starting from an assumed initial period), with no stabilising satellite.

In the Hadean context — a T-Tauri Sun whose coronal mass ejections are 10 to 100 times more intense than today's (Feigelson & Montmerle, 1999; Airapetian et al., 2016), a dense silicate atmosphere at 2,000–4,000 K acting as a thermal blanket (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), continuous planetesimal bombardment (Agnor et al., 1999; Kokubo & Genda, 2010), and active short-lived radioactivity (Urey, 1955; Huss et al., 2009) — the proto-Earth's axis undergoes large-amplitude free nutation within $[40°, 70°]$ in its own co-rotating frame (Laskar et al., 1993a; Laskar, 1993b; Touma & Wisdom, 1993; Li & Batygin, 2014). This is *not* an astronomical obliquity relative to a non-existent Hadean ecliptic; it is a purely internal geometric oscillation of the fluid body.

In this setting, a single engine — the progressive segregation of Fe-Ni toward the forming core — drives three coupled transitions. The first is a rheological transition: the silicate magma acquires a Bingham-Herschel yield stress (Bingham, 1922; Herschel & Bulkley, 1926) and organises a Coherent Magmatic Torus (CMT) within the intertropical band $|\phi| < 30°$. The second is a mechanical transition: the CMT undergoes two to three episodes of cohesive hypersonic ejection beyond the Roche radius, governed by a double-well potential and the stochastic Kramers crossing rate (Kramers, 1940). Each ejection alters the proto-Earth's inertia tensor and shifts the instantaneous rotation axis to a new obliquity value within $[40°, 70°]$; since the intertropical band is defined perpendicular to this axis, this realignment displaces the active band itself, so that the next torus re-forms at a different location and on a spheroid whose own shape has changed — each successive CMT is therefore geometrically distinct from the previous one, underlying the observed nearside/farside crustal dichotomy. The third is a magnetic transition: the terrestrial dynamo switches on $\approx$ 350 Myr after ejections cease, consistent with paleomagnetic data from Jack Hills zircons (Tarduno et al., 2025).

The central equation gives an ejected mass of 2.2–4.0 $\times$ 10^{22} kg per episode; with $N = 2$–3 episodes and a capture efficiency of 0.70, the lunar mass is reconstituted. The mechanism requires no external impactor and satisfies the constraints of isotopic identity, iron depletion, and angular momentum as direct mechanical consequences. The central prediction is the existence of one or more seismic interfaces between 200 and 530 km depth, with an impedance contrast $|R| \in [0.01, 0.04]$. This prediction is testable by **Chang'e 7** (South Pole, August 2026), **FSS** (Farside Seismic Suite), **LEMS** (Lunar Environment Monitoring Station), and **Artemis III** (2028–2029). Preliminary observational support comes from Chang'e-6 samples (Yue et al., 2026; Xu et al., 2025): norites dated at 4247 ± 5 Myr (consistent with Episode 1) and anomalously high Fe/Mn ratios in deep olivines, both consistent with the predicted Fe/Si gradient (P3) and the South Pole Fe/Si anomaly (P6). The confinement of the CMT to the intertropical band admits an effective description via a nonlinear Schrödinger equa-

1 Introduction: why a new theory of lunar formation?

1.1 The giant-impact problem

The canonical giant-impact model (Cameron & Ward, 1976; Hartmann & Davis, 1975) dominated planetary science for nearly fifty years. In this scenario, a Mars-sized body (Theia) struck the proto-Earth, and the resulting debris disc accreted to form the Moon. This model explains several first-order features: the angular momentum of the Earth–Moon system, the relatively modest size of the lunar core, and the absence of a massive atmosphere.

However, over the past decade, growing geochemical and chronological difficulties have eroded confidence in the giant-impact scenario as a complete explanation:

1. **Earth–Moon isotopic identity**: the isotopic compositions of oxygen, titanium, tungsten, and chromium are nearly identical between Earth and the Moon to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The giant-impact model requires either that Theia had an isotopic composition identical to Earth's (an improbable coincidence at $< 1\%$), or complete post-impact isotopic mixing (Canup, 2012), which is not naturally predicted by hydrodynamic simulations.
2. **Iron depletion**: the Moon is significantly depleted in siderophile elements relative to Earth's mantle (O'Neill, 1991; Jones & Drake, 1986). The giant impact explains this depletion by assuming that Theia was pre-differentiated (with a metallic core that accreted to Earth), but this assumption adds a free parameter.
3. **Crustal dichotomy**: the lunar farside is thicker (≈ 54 km) and older than the nearside (≈ 34 km) (Wieczorek et al., 2013). The giant impact does not predict this asymmetry.
4. **Terrestrial dynamo delay**: Earth's magnetic field is detected in Jack Hills zircons at 4.2 Ga (Tarduno et al., 2025), some ≈ 350 Myr after the estimated formation of the Moon (4.55 Ga). The giant impact offers no explanation for this delay.

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in lunar formation (Sossi et al., 2025). This conclusion motivates the exploration of alternatives grounded in the internal dynamics of the proto-Earth.

1.2 The Triple Phase Transition: an endogenous alternative

This work proposes that the Moon did not form through an external impact, but through an endogenous process within the differentiating proto-Earth itself. The central thesis is as follows.

Hypothesis H0 (central thesis). The Moon formed through a series of *three coupled transitions* — rheological, mechanical, and magnetic — triggered by the single engine of Fe-Ni segregation toward the forming core. The result is a Moon built layer by layer in two to three episodes of cohesive magmatic ejection, each layer archiving the state of Hadean differentiation at the moment of its accretion.

The mechanism rests on a necessary thermodynamic observation: every Earth-mass planet emerges from accretion in a state of near-total silicate-mantle melting (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). Accretion energy exceeds the total melting energy by a factor of ≈ 155. The proto-Earth is therefore a rapidly rotating magmatic body ($T_{\rm rot} \approx 3.5$ h) with no stabilising satellite.

Within this rapidly rotating fluid body, the rotation axis undergoes large-amplitude free nutation within $[40^\circ, 70^\circ]$ in the co-rotating frame (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014) — not an astronomical obliquity, but an internal oscillation of the fluid body. This oblique tilt is the geometric condition that allows the radial Coriolis force ($f_{\rm rad} = 2\Omega U \sin \varepsilon$) to destabilise the flow rather than confine it.

1.3 Organisation of the manuscript

The manuscript is organised as follows. Section 2 presents the Hadean physical context (rotation, obliquity, silicate atmosphere). Section 5 presents the intertropical band as the ejection site. Section 6 addresses the Taylor-Proudman objection. Section 7 inventories the forces. Section 8 explicitly ranks all hypotheses. Section 9 presents the three coupled transitions (rheological, mechanical, magnetic) with the complete resolution of the double well. Section 10 presents the mass budget. Section 11 presents angular-momentum transport. Section 16 gives the 3–4 week chronology. Section 17 presents the internal stratification. Section 18 discusses observational constraints and existing evidence. Section 19 presents the falsifiable predictions. Section 20 presents the validation windows. Section 21 discusses the main objections. Section 22 presents the limitations. Section 23 concludes.

2 The Hadean inferno: physical context

Before turning to the physics of the Triple Phase Transition, it is essential to establish a structural fact, not merely a descriptive one, about the setting in which it operates: no stationary state is physically accessible within it. This is not simply an agitated period among others in Solar System history — it is a regime in which every process that might, on its own, return the system toward equilibrium is itself perturbed, on a timescale shorter than its own relaxation time, by at least one other independent process. It is this structural absence of rest, rather than an inventory of individual circumstances, that this section sets out to establish: it conditions directly the reading of Section 3 (the axial oscillation is never damped for lack of a stabiliser) and, further on, that of Section 9.2 (each ejection episode occurs not on a system at rest, but on a system already and continuously out of equilibrium).

2.1 The dynamical context of terminal accretion

The Hadean period (4.568–4.4 Ga) superimposes a set of physical processes whose characteristic timescales overlap without any one of them dominating long enough to impose a stable regime:

- **Continuous planetesimal accretion**: the protoplanetary disc remains populated with planetesimals and embryos, with each major impact occurring every 10^3–10^5 years (Agnor et al., 1999; Kokubo & Genda, 2010) — an interval far shorter than the characteristic viscous relaxation time of the fluid body (Section 3), so that the effect of one impact never has time to fade before the next.
- **Rapid rotation**: the proto-Earth rotates with a period $\approx$ 3.5 h, a regime in which Coriolis and centrifugal effects dominate the internal dynamics and amplify any perturbation rather than damping it.
- **Absence of a tidal stabiliser**: with no Moon, obliquity is free to evolve over a wide chaotic range, with no restoring mechanism toward an equilibrium value.
- **A young, hyperactive Sun**: T-Tauri-type stars produce flares and coronal mass ejections (CMEs) at a frequency reaching 0.2–0.4 events per day for superflares with confirmed mass ejection, and over one event per day for white-light flares alone, based on direct observations of young solar-type stars ($\sim$ 30–125 Myr) (Namekata et al., 2026), compared with roughly one superflare per few thousand years for the present-day Sun (Feigelson & Montmerle, 1999; Airapetian et al., 2016) — a stochastic forcing whose characteristic timescale is daily, where that of planetesimal impacts is millennial and that of free nutation likewise on the order of a day (Section 3): these three clocks never synchronise, they collide.
- **Short-lived radioactivity**: the isotopes ^{26}Al and ^{60}Fe remain active, continuously injecting heat into the mantle (Urey, 1955; Huss et al., 2009), with no thermal resting phase.
- **Dense silicate atmosphere**: a rock-vapour envelope at $T \sim 2{,}000$–$4{,}000$ K and $P \sim 10^5$–10^7 Pa constitutes a thermal blanket and a lateral confining agent (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), but it stabilises nothing by itself: it extends the window during which the preceding processes can continue to act.

It is not the list of these processes that matters, but their simultaneity. Taken separately, each could in principle average out over time and allow a quasi-stationary behaviour to emerge; taken together, at frequencies that are commensurable on no common timescale, they structurally preclude such a regime from settling in. The T-Tauri Sun, although it provides one of the most frequent forcings, is therefore not the cause of the Hadean chaos: it is only one contributor among others, whose combination renders any equilibrium inaccessible. It is in this already and continuously out-of-equilibrium context that the ejection episodes themselves will occur (Section 9.2): not as isolated perturbations striking an otherwise calm system, but as additional shocks added to a system that, at no point in its active history, ever truly came to rest.

2.2 The silicate atmosphere: thermal blanket and lateral confinement

At $t = 0$, the proto-Earth is enveloped in a dense silicate atmosphere — rock vapour, gaseous SiO, Mg, and Fe — at temperatures of 2,000 to 4,000 K and pressures of 10^5–10^7 Pa. This cocoon plays two critical roles in the TPT.

First role: thermal insulation. The silicate atmosphere has strong infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). Without this atmosphere, the proto-Earth would have cooled within $\sim 10^4$ years. With it, the melting window extends to 30–50 Myr — precisely the timescale of core formation (Kleine et al., 2002; Yin et al., 2002). This is not an ad hoc assumption: it is the physical consequence of a rock-vapour atmosphere at $2,000$–$4,000$ K. The same mechanism operates in magma-ocean models (Elkins-Tanton, 2008; Lebrun et al., 2013).

Second role: lateral confinement. The confining pressure ($P \approx 10^5$–10^7 Pa) opposes lateral expansion of the CMT flow during ejection. It guarantees the cohesion of the ejected sheet. The high sound speed in this scorching atmosphere ($c_{\text{sound}} \approx 1,500$ m s^{-1} at $T_{\text{atm}} \approx 3,000$ K) reduces the effective Mach number of the flow, improving cohesion relative to ejection into vacuum.

Without this atmosphere, the melting window would close within $\sim 10^4$ years. With it, the window extends to 30–50 Myr (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), matching the timescale of core formation (Kleine et al., 2002; Yin et al., 2002).

2.3 Continuous planetesimal bombardment

The protoplanetary disc remains populated with planetesimals and embryos throughout the active TPT window (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2001). Each impact: (1) reinjects thermal energy into the mantle, sustaining melting; (2) generates a spherical pressure wave propagating through the entire melt pool, perturbing $U(r)$ and stochastically contributing to Kramers barrier crossings; (3) delivers an impulsive torque on the spin vector, resetting the axial oscillation.

2.4 Short-lived radioactivity: internal heat source

The isotopes ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (half-life 2.6 Myr) remain active during the first million years (Urey, 1955; Huss et al., 2009), injecting $\approx 3 \times 10^{29}$ J of additional heat into the mantle and contributing to sustained melting throughout the TPT window.

3 Hadean obliquity $\varepsilon \in [40^\circ, 70^\circ]$: a fundamental geometric clarification

The interval $\varepsilon \in [40^\circ, 70^\circ]$ is not a free parameter. It is constrained by five independent and convergent physical arguments. Before listing them, a fundamental geometric clarification is required.

Geometric clarification. In this work, ε denotes the instantaneous angle between the rotation vector $\mathbf{\Omega}(t)$ and the principal axis of inertia of the Maclaurin spheroid, **defined in the body's co-rotating frame**. It is *not* the astronomical obliquity relative to the ecliptic plane: that concept has no physical meaning for the Hadean proto-Earth, which possesses neither a fixed reference ecliptic nor a Moon. It is a purely internal geometric angle describing the *oscillation* — the free nutation — of the rotation axis relative to the body's own shape.

Definition of the intertropical band. The Hadean intertropical band $|\phi| < 30^\circ$ is defined perpendicular to the *instantaneous* rotation axis $\mathbf{\Omega}(t)$, not relative to a fixed geographic frame. The $\phi = 0$ plane is therefore the body's *instantaneous dynamical equator*, which oscillates with the free nutation of $\mathbf{\Omega}(t)$ — it is not a fixed geographic equator. The coordinate ϕ is the oblique latitude measured from this instantaneous equator. This is the plane where the radial Coriolis force $f_{\text{rad}} = 2\Omega U \sin \varepsilon$ is maximal and where the effective gravity $g_{\text{eff}}(\phi)$ is minimal (Section 4.3). The band is thus a dynamical structure tied to the instantaneous equator, not a fixed latitude. This identification of ϕ with the instantaneous dynamical equator, rather than a fixed geographic equator, is used directly in Section 7.3 and again in the derivation in Appendix C.

It follows directly from the Solar System still lacking any external reference plane — no ecliptic, no stable lunar orbit, no fixed solar axis — that the very notion of an "intertropical" band is not geographic but purely relative, defined by this internal axis rather than by a solar or ecliptic frame that does not yet exist. Throughout this work, any term evoking ordinary planetary geography ("intertropical", "equator", "latitude") should be read as a formal analogy with the geometry of a rotating spheroid, not as a reference to an astronomy that has not yet come into being at this epoch.

Argument 1 — Absence of a tidal stabiliser (fundamental). The proto-Earth has no Moon. Without a massive satellite, there is no lunar tidal torque available to damp the precession of the spin axis and keep it near a stable equilibrium. This argument is logically prior to all others: it is precisely the formation of the Moon that we seek to explain here, so invoking it as a stabiliser would be circular. The absence of a tidal stabiliser is therefore a necessary, not contingent, condition of the Hadean proto-Earth.

Argument 2 — Laskar et al. (1993): a chaotic zone with no satellite. Laskar et al. (1993a) showed that without its Moon, Earth's obliquity would evolve chaotically over a zone extending from near $0°$ to $\approx 85°$. Although this result strictly applies to a present-day Earth rotating in ≈ 24 h, it establishes the general principle: *the Moon is the stabiliser, and without it, large obliquity excursions are the norm*. For the Hadean proto-Earth, this principle applies *a fortiori*.

Argument 3 — Rapid rotation at $T_{\rm rot} = 3.5$ h. The precession constant satisfies $\alpha_{\rm prec} \propto \omega$. At $T_{\rm rot} = 3.5$ h, the precession frequency is $\approx 7\times$ higher than at 24 h, pushing the secular spin-orbit resonances identified by Laskar et al. (1993a) out of reach. Li & Batygin (2014), working on a rapidly rotating moonless Earth, found that obliquity remains chaotic but *diffuses slowly* over a wide range. Rapid rotation does not stabilise the axis — it merely changes the nature of the chaos, from resonant to diffusive.

Argument 4 — Continuous planetesimal bombardment. Every major impact during terminal accretion (Agnor et al., 1999) abruptly resets the spin vector. For a low-viscosity fluid body ($\eta \sim 10^{1\text{–}3}$ Pa s), the viscous relaxation time toward axial alignment, of the form $\tau \sim \eta/(\rho g R_{\rm eq,proto})$ (Goldreich & Mitchell, 2010), is extremely short ($\ll 1$ yr) compared with the interval between major impacts (10^3–10^5 yr): $\tau_{\rm relax} \ll \tau_{\rm pert}$, where $\tau_{\rm pert} \sim 10^3$–$10^5$ yr is the typical interval between successive major impacts (Agnor et al., 1999). Misalignment is continuously maintained by the frequency of impacts.

Argument 5 — T-Tauri CME torques. As described in Section 7.4, the impulsive torques from CMEs act on timescales of days to weeks, comparable to $\tau_{\rm pert}$, and constitute a continuous stochastic source of obliquity perturbation, in addition to impacts (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

The interval $\varepsilon \in [40°, 70°]$ is not an ad hoc assumption. It is the predictable consequence of the dynamics of a rapidly rotating fluid body, lacking a tidal stabiliser, subject to continuous stochastic perturbations. The simulations of Laskar et al. (1993a) for a moonless Earth, and of Li & Batygin (2014) for a rapidly rotating moonless Earth, converge on this interval — the same one adopted by Laskar et al. (1993a) and Touma & Wisdom (1993). In this oblique geometry, the Coriolis force decomposes into radial and horizontal components (Section 7.3). The radial component $f_{\rm rad} = 2\Omega \sin\varepsilon \cdot U$ is maximal for $\varepsilon \in [40°, 70°]$.

Section synthesis

Five convergent arguments support $\varepsilon \in [40°, 70°]$. The absence of a tidal stabiliser is logically necessary. Arguments 2 through 5 provide independent physical justification for a wide and sustained misalignment. The interval $[40°, 70°]$ is adopted as the physically motivated Hadean interval; the mechanism operates for any $\varepsilon > 45°$ (Section 7.3). The intertropical band $|\phi| < 30°$ is, throughout this work, tied to the instantaneous dynamical axis $\mathbf{\Omega}(t)$, not to a fixed geographic frame.

3.1 Relaxation time and rheological reinforcement

The viscous relaxation time toward axial alignment for an isolated fluid body, $\tau_{\text{relax}} \sim \eta/(\rho g R_{\text{eq,proto}})$ (Goldreich & Mitchell, 2010), is very short ($\ll 1$ yr) at low viscosity, as established in Section 3. Bingham-Herschel rheology locally slows this relaxation. The rheological response time to a stress perturbation is:

$$\tau_{\text{BH}} \sim \frac{\tau_y}{\mu \dot{\gamma}_{\text{typ}}} \approx 10^2\text{–}10^4\,\text{s} \quad (\tau_y \sim 10^2\text{–}10^4\,\text{Pa},\ \mu \sim 1\text{–}100\,\text{Pa s},\ \dot{\gamma}_{\text{typ}} \sim 1\,\text{s}^{-1})\text{(Giordano et al., 2008)}, \tag{1}$$

where μ is dynamic viscosity and $\dot{\gamma}_{\text{typ}}$ the typical shear rate. This time remains far shorter than the interval between impacts τ_{pert}, which means that mass redistribution is locally frozen by the yield stress between perturbations, delaying local shape relaxation without affecting the conclusion of Argument 4: it is the frequency of perturbations, not an intrinsically long relaxation time, that maintains axial misalignment.

4 Initial state of the proto-Earth

4.1 Near-total melting: a necessary thermodynamic consequence

The formation of an Earth-mass planet releases a considerable amount of gravitational energy. In the homogeneous-sphere approximation (Safronov, 1969), the total accretion energy is:

$$E_{\text{grav}} = \frac{3\,G\,M_\oplus^2}{5\,R_{\text{eq,proto}}} \approx 2.49 \times 10^{32}\,\text{J}, \tag{2}$$

where G is the gravitational constant, $M_\oplus$ Earth's mass, and $R_{\text{eq,proto}} \approx 7.4$ Mm the equatorial radius of the undifferentiated proto-Earth. This value is consistent with the commonly cited gravitational binding energy of Earth, $\approx 2.5 \times 10^{32}$ J (Birch, 1965; Flasar & Birch, 1973).

Equation (2) is a thermodynamic bound, not a model of instantaneous formation. It computes the *total* gravitational energy released during accretion, independent of its duration. This is the standard approach in the literature (Solomatov, 2000;

Elkins-Tanton, 2012; Rubie et al., 2015). The manuscript explicitly cites Kleine et al. (2002); Yin et al. (2002) as evidence that accretion lasted ~ 30 Myr.

The energy required to melt the entire silicate mantle (Stixrude & Lithgow-Bertelloni, 2014):

$$E_{\text{fus}} \approx M_{\text{mantle}} \times L_{\text{fus}} \approx 4 \times 10^{24}\,\text{kg} \times 4 \times 10^{5}\,\text{J}\,\text{kg}^{-1} \approx 1.6 \times 10^{30}\,\text{J}, \tag{3}$$

where M_{mantle} is the mass of the silicate mantle and L_{fus} the specific latent heat of fusion.

The resulting energy ratio is:

$$E_{\text{grav}} / E_{\text{fus}} \approx 155. \tag{4}$$

This precise value follows from the parameters adopted here ($R_{\text{eq,proto}} \approx 7.4$ Mm, $M_{\text{mantle}} \approx 4 \times 10^{24}$ kg, $L_{\text{fus}} \approx 4 \times 10^{5}\ \text{J}\,\text{kg}^{-1}$), explicitly stated above; it is not a universal constant of Earth. Other, equally defensible parameter choices in the literature — a radius close to Earth's present radius, a smaller mantle mass, or a higher latent heat — yield different numerical values, typically between ≈ 80 and ≈ 190 depending on the assumptions adopted. What remains robust across this entire range, and what constitutes the consensus result in the literature (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015), is the qualitative conclusion: accretion energy exceeds the melting energy of the silicate mantle by more than two orders of magnitude, making near-total melting of the proto-Earth a robust thermodynamic consequence rather than a result contingent on a particular choice of parameters.

Why melting was sustained despite radiative cooling

Accretion lasted ~ 30 Myr (Kleine et al., 2002; Yin et al., 2002). During this period, radiative cooling occurred. Four independent mechanisms explain why melting was sustained despite this cooling.

Mechanism 1 — The silicate atmosphere as a thermal blanket. The dense silicate atmosphere ($P \approx 10^{5}$–10^{7} Pa, $T \approx 2{,}000$–$4{,}000$ K) has strong infrared opacity. It reduces surface radiative cooling by a factor of 10^{4} to 10^{5} (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). The effective cooling time shifts from $\sim 10^{4}$ years to 30–50 Myr.

Mechanism 2 — Additional heat sources. Four heat sources extend melting beyond the initial accretion energy:

- **Fe-Ni segregation heat:** $\approx 3 \times 10^{30}$ J (Flasar & Birch, 1973; Rubie et al., 2003).
- **Short-lived radioactivity:** ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (half-life 2.6 Myr) inject $\approx 3 \times 10^{29}$ J (Urey, 1955; Huss et al., 2009).
- **Continuous planetesimal bombardment:** reinjection of thermal energy (Agnor

et al., 1999; Kokubo & Genda, 2010).

- **T-Tauri irradiation:** maintains surface temperatures above 2,000 K (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

Mechanism 3 — The TPT does not require melting of the entire mantle. The Triple Phase Transition requires melting only within the Hadean intertropical band $|\phi| < 30°$ (upper and mid mantle). The factor of 155 thermodynamically guarantees this partial melting, even if the deep mantle was not entirely molten (Nomura et al., 2011).

Mechanism 4 — The TPT active window is 30–50 Myr. The TPT occurs *within* the 30–50 Myr window during which melting is sustained by the mechanisms above. Core formation over $\sim$ 30 Myr (Kleine et al., 2002; Yin et al., 2002) is later than, or concurrent with, this window — this is not a contradiction.

Near-total melting constitutes the mandatory starting point. The objection based on the $\sim$ 30 Myr accretion timescale does not contradict the mechanism: the TPT occurs *within* the melting window sustained by the silicate atmosphere, the additional heat sources, and continuous bombardment.

Hypothesis H1 — Total volumetric melting. At the onset of Fe-Ni segregation, the entire silicate mantle lies above the liquidus, with no stable internal solid interface during the active window 4.568–4.4 Ga. This is a self-gravitating fluid volume, not a magma ocean resting on a substrate: the dynamics, rheology, and response to perturbations are physically distinct.

Caveat: Nomura et al. (2011) suggest that the deep lower mantle might not have been entirely molten, owing to the rise of the solidus with pressure. The TPT requires melting only within the Hadean intertropical band $|\phi| < 30°$ (upper and mid mantle), which the factor of 155 thermodynamically guarantees. The adiabatic temperature profile (Appendix A) confirms melting throughout the mantle.

4.2 Initial rotation period: $T_{\mathrm{rot}} \approx 3.5$ h

The proto-Earth's initial rotation period is not a free parameter: it is constrained directly by the terrestrial-planet formation literature at $T_{\mathrm{rot}} \approx 3$–$5$ h.

1. T-Tauri magnetic torque. The young, rapidly rotating Sun ($\approx$ 3 days, Bouvier et al. 2014) couples magnetically to the inner protoplanetary disc (Königl, 1991; Shu et al., 1994), transferring angular momentum to inner-orbit matter and accelerating the proto-Earth.

2. N-body simulations of terrestrial planet formation. N-body simulations show that final spin periods of Earth-mass planets lie in the 3–5 h range (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2013). Agnor et al. (1999) find that protoplanetary embryos rotate rapidly during the final stages of accretion, upon their first collision, suggesting that the proto-Earth may have rotated rapidly before any subsequent major impact. Kokubo & Genda (2010), using N-body simulations of protoplanet formation under realistic accretion conditions, find final rotation periods of 3–5 h for Earth-mass bodies, consistent with Chambers (2013). The value $T_{\mathrm{rot}} \approx 3.5$ h adopted here falls within this independently established interval, and is of the same order of magnitude as the fast pre-impact rotations considered in the giant-impact models of Ćuk & Stewart (2012) (2.3–2.7 h) and Canup (2012) (a 3 h pre-impact rotation in one simulated case) — although neither specific value is identical to 3.5 h, nor does the TPT rely on the giant-impact hypothesis itself.

Why rapid rotation requires an acceleration mechanism. Dones & Tremaine (1993) show analytically that orderly accretion from a uniform or slowly varying disc of small bodies cannot produce rotation as rapid as that of Earth or Mars, regardless of the disc's velocity dispersion: the prograde and retrograde contributions of individual planetesimals cancel out nearly by symmetry. The rapid spin velocities of terrestrial planets are therefore most naturally explained by one or a few collisions between large bodies late in accretion — a requirement that the TPT satisfies through the continuous planetesimal bombardment of Section 2, regardless of whether a single dominant impactor is invoked (as in the giant-impact hypothesis). Consistent with this, Kokubo & Ida (2007), using N-body simulations of the embryo-merging phase of accretion, find that planetary spin angular velocities follow a Maxwellian distribution with substantial spread: there is no single preferred rotation period predicted by accretion physics alone, only a wide statistical range within which $T_{\mathrm{rot}} \approx 3.5$ h is a plausible value, not a unique result derivable from accretion physics alone.

3. Skater effect: Fe-Ni segregation. The migration of Fe-Ni toward the forming core reduces the moment of inertia as segregation proceeds. With $I_f/I_0 \approx 0.82$ (Rubie et al., 2015), this contributes an additional acceleration of about 18% to rotation over the course of the TPT window, in addition to the rotation velocity already acquired by the proto-Earth during accretion. This effect is qualitative and contributory: it is not used here to derive $T_{\mathrm{rot}} \approx 3.5$ h from an assumed pre-segregation value, since no such value is independently constrained.

4. Distinction from the synestia regime. Lock & Stewart (2017) define the corotation limit (CoRoL) as the rotation rate at which a body's equatorial velocity equals the local Keplerian orbital velocity; bodies exceeding this limit form a distended, partially-vaporised structure they term a synestia, rather than a simple rigidly rotating body. For the proto-Earth parameters adopted here ($R_{\mathrm{eq,proto}} \approx 7.4$ Mm), the corotation limit corresponds to a period of about 1.8 h. At $T_{\mathrm{rot}} \approx 3.5$ h, Ω reaches only $\approx 50\%$ of this limit: the proto-Earth described by the TPT therefore lies well within the coherent rigid-body rotation regime, not within a synestia.

On the initial rotation period

The value $T_{\text{rot}} \approx 3.5$ h is constrained directly by the terrestrial-planet formation literature (Agnor et al. 1999; Kokubo & Genda 2010; Chambers 2013; 3–5 h), is of the same order of magnitude as the fast rotations considered in giant-impact models (Ćuk & Stewart, 2012; Canup, 2012) without depending on the giant-impact hypothesis, and is consistent with the theoretical requirement that rapid terrestrial spin requires an acceleration mechanism late in accretion (Dones & Tremaine, 1993), within the wide statistical range found for planetary spin in N-body simulations (Kokubo & Ida, 2007). It is not derived from an assumed initial period via a contraction argument. The Coriolis-compensation parameter b', a simple ratio between the critical ejection velocity U_{crit} and the CMT's geostrophic equilibrium velocity, provides an independent cross-check: Fe-Ni segregation increases b', ejections decrease it, and the equilibrium point $b' = 1$ is a stable attractor that naturally selects $T_{\text{rot}} \approx 3.5$ h and $\varepsilon \approx 55°$. The complete derivation is given in Appendix B.

4.3 Geometry of the proto-Earth: the Maclaurin spheroid

A rapidly rotating fluid mass adopts the shape of a Maclaurin spheroid (Chandrasekhar, 1969). The equatorial radius is $R_{\text{eq,proto}} \approx 7.4$ Mm, combining reduced density (+4%), thermal expansion (+7.5%), and equatorial bulging (+7.5%). The dynamical flattening, given by $J_2 = q/[2\Phi(\bar{C}) - 3]$ where $q = \Omega^2 R^3_{\text{eq,proto}}/(GM_\oplus)$ is the rotation parameter and $\bar{C}$ the normalised axial moment of inertia (Murray & Dermott, 1999), reduces to $J_2 \approx q/2$ for a strictly homogeneous body ($\bar{C} = 0.4$):

$$J_2 \approx \frac{1}{2} \cdot \frac{\Omega^2 R^3_{\text{eq,proto}}}{GM_\oplus} \approx 0.13, \tag{5}$$

a value consistent with the range $J_2 \sim 0.10$–0.13 expected for a body close to homogeneity (Murray & Dermott, 1999), giving an equatorial bulge $\delta R \approx 600$–650 km. Effective gravity varies with latitude ϕ (measured from the instantaneous dynamical equator, as defined in Section 3) according to:

$$g_{\text{eff}}(\phi) = g^{(0)}_{\text{eff}} \left(1 + \tfrac{3}{2} J_2 \sin^2\phi\right) - \Omega^2 \, r(\phi) \cos^2\phi, \tag{6}$$

where $r(\phi)$ is the local radius. Effective gravity is minimal within the band $|\phi| < 30°$: this is the geometric attractor of the CMT.

The free (Euler) nutation period, $\omega_E = \Omega(C - A)/A$ where C and A are the polar and equatorial moments of inertia of the spheroid (Murray & Dermott, 1999), with $(C - A)/A \equiv f \approx 0.075$ for a body close to homogeneity at the dynamical flattening factor J_2 adopted here (Eq. 5), is:

$$\omega_E = \Omega f \approx 3.74 \times 10^{-5}\,\mathrm{rad\,s^{-1}}, \qquad T_{\mathrm{nut}} \approx 46.7\,\mathrm{h}. \tag{7}$$

This frequency plays a central role in the parametric elliptical instability (Section 7.6).

Stability of the Maclaurin spheroid at this rotation rate. A rapidly rotating self-gravitating fluid body is not necessarily constrained to adopt the axisymmetric Maclaurin shape: beyond a critical rotation rate, the Maclaurin sequence bifurcates toward the triaxial Jacobi sequence. The bifurcation criterion compares two characteristic timescales: the self-gravitation timescale, $t_g \sim 1/\sqrt{G\bar{\rho}}$, and the rotation period, $t_{\mathrm{rot}} = 2\pi/\Omega$; their ratio defines the dimensionless number $\Omega^2/(\pi G\bar{\rho})$. This criterion is derived analytically by Chandrasekhar (1969) (Chapter 4, complete derivation; see also Chapters 3–5 for Maclaurin spheroids and Chapters 6–7 for Jacobi ellipsoids and the bifurcation itself) as the point at which the family of Jacobi ellipsoids branches off from the Maclaurin sequence — not as an empirical approximation — and is presented from a historical perspective by Jeans (1929), revisited in the stellar context by Tassoul (1978), the theory being formally identical for any self-gravitating fluid; Lamb (1932) gives the classical hydrodynamic treatment, and Durand (1964) the ellipsoidal integrals on which Chandrasekhar's derivation rests. This bifurcation occurs at $\Omega^2/(\pi G\bar{\rho}) \approx 0.374$, with Maclaurin spheroids becoming genuinely unstable only beyond $\Omega^2/(\pi G\bar{\rho}) \approx 0.440$.

The ρ in this criterion is the uniform density of the homogeneous fluid in the theoretical model. Its application to a real planetary body substitutes for this uniform density the mean density $\bar{\rho} = M/(\frac{4}{3}\pi R^3)$, an approximation explicitly discussed by Chandrasekhar (1969) (Chapters 1 and 5) and Tassoul (1978) (Chapter 2). For the Hadean proto-Earth, $\bar{\rho}$ is not directly measurable: it depends on the degree of differentiation already attained, which is itself not precisely known. Available estimates place the mean density of a homogeneous Earth of the same mass, prior to complete core segregation, around $4,100$–$4,500\ \mathrm{kg\,m^{-3}}$, against $5,510\ \mathrm{kg\,m^{-3}}$ for the present-day fully differentiated Earth (Rubie et al., 2015; Trønnes et al., 2019; Rubie et al., 2016; Hirose et al., 2021). These values are themselves derived from mineralogical models (chondritic or pyrolitic composition, equations of state) rather than direct measurement, and therefore constitute a modelling range rather than a physical constant.

For the proto-Earth parameters adopted here ($R_{\mathrm{eq,proto}} \approx 7.4$ Mm, mean density $\bar{\rho} \approx 4,300\ \mathrm{kg\,m^{-3}}$, the median value of this range, for a still weakly differentiated body), $T_{\mathrm{rot}} \approx 3.5$ h gives $\Omega^2/(\pi G\bar{\rho}) \approx 0.276$ — below the Jacobi bifurcation point (0.374), with a margin of about 26%. At this density, the bifurcation threshold itself corresponds to $\Omega_c = \sqrt{0.374\,\pi G\bar{\rho}} \approx 5.8 \times 10^{-4}\,\mathrm{rad\,s^{-1}}$, i.e. $T_{\mathrm{rot}} \approx 3.0$ h, and the strict dynamical-instability threshold to $T_{\mathrm{rot}} \approx 2.8$ h. The axisymmetric Maclaurin geometry adopted throughout this work is therefore consistent with the TPT rotation rate, and does not require invoking a triaxial or otherwise more complex equilibrium figure; the available margin nevertheless remains sensitive to the exact value of $\bar{\rho}$ adopted, which is not known with a precision better than the modelling range cited above.

5 The intertropical band $|\phi| < 30°$: the ejection site

A fundamental and often misunderstood point: the CMT is not an equatorial structure. It is a coherent zonal band spanning $\pm 30°$ of oblique latitude — roughly one third of the proto-Earth's surface. This extent results from three independent and convergent mechanisms.

As established in Section 3, the band $|\phi| < 30°$ is defined perpendicular to the instantaneous rotation axis $\mathbf{\Omega}(t)$, not relative to a fixed geographic frame. This is the plane where the radial Coriolis force is maximal and where effective gravity is minimal.

Mechanism 1 — Maclaurin geometry. Effective gravity is minimal within the band $|\phi| < 30°$ (Eq. 6). Molten material accumulates there through centrifugal effects, independent of any other argument. This is a consequence of the ellipsoidal shape imposed by rapid rotation.

Mechanism 2 — The geostrophic inverse cascade. The CMT's Rossby number, with $U_{\text{turb}} \approx 0.8\ \text{km}\,\text{s}^{-1}$ (Eq. 19), is:

$$\text{Ro} = \frac{U_{\text{turb}}}{2\,\Omega\,R_{\text{eq,proto}}} \approx 0.108 \ll 1, \tag{8}$$

In this regime, turbulent energy cascades toward large scales (inverse cascade) (Rhines, 1975). The Rhines scale:

$$L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}}, \qquad \beta = \frac{2\Omega\cos\varepsilon}{R_{\text{eq,proto}}}, \tag{9}$$

gives $L_\beta \approx 0.52\,R_\oplus$ — corresponding to harmonic mode $\ell = 1$, $m = 1$: the largest coherent structure possible on a sphere. The CMT is the statistical attractor of this cascade (Rhines, 1975; Sukoriansky et al., 2007).

Mechanism 3 — The radial Coriolis force. $f_{\text{rad}} = 2\Omega U \sin\varepsilon$ concentrates the flow within the oblique intertropical band and keeps it coherent there.

Section synthesis

These three mechanisms — Maclaurin geometry, geostrophic inverse cascade, and radial Coriolis force — are independent and converge on the same band $|\phi| < 30°$. Their simultaneous convergence is an argument for the structural robustness of the theory.

6 The CMT as a purely toroidal flow: response to the Taylor-Proudman objection

Any geophysical fluid dynamicist confronted with the CMT will naturally raise the following objection: in a rapidly rotating fluid (Ro $\ll 1$), the Taylor-Proudman theorem imposes $(\boldsymbol{\Omega} \cdot \nabla)\boldsymbol{u} = 0$, generating columnar structures parallel to $\boldsymbol{\Omega}$, rather than a single oblique band $|\phi| < 30°$. The objection is legitimate and deserves a precise answer.

6.1 The poloidal/toroidal decomposition

Any velocity field within a spherical shell decomposes into a poloidal part (radial and meridional components) and a toroidal part (purely azimuthal component):

$$\boldsymbol{u} = \boldsymbol{u}_{\text{pol}} + \boldsymbol{u}_{\text{tor}}, \qquad \boldsymbol{u}_{\text{tor}} = \nabla \times (\psi \hat{\boldsymbol{r}}).$$

The Taylor-Proudman theorem, in its classical form, constrains the poloidal field (which carries vorticity along $\boldsymbol{\Omega}$) and produces Taylor columns. It does not constrain the purely toroidal field.

For a purely azimuthal flow independent of longitude ($u_r = u_\theta = 0$ on average, $u_\phi = u_\phi(r, \phi)$), the Taylor-Proudman constraint is automatically satisfied:

$$(\boldsymbol{\Omega} \cdot \nabla) u_{\text{tor}} = 0 \tag{10}$$

for any toroidal field, with no additional condition. The balance is cyclostrophic, not geostrophic: the centrifugal pressure gradient balances the Coriolis force within the azimuthal plane.

6.2 Why curved columns do not extend beyond the active band

In an obliquely rotating spherical shell ($\varepsilon \neq 0$), Taylor columns are curved along the projected field lines of $\boldsymbol{\Omega}$. They tend to channel flow away from its source — unless a cut-off mechanism stops them.

This is precisely the role of the yield-stress threshold τ_y. In a Bingham-Herschel fluid, flow propagates only in regions where the stress exceeds τ_y. As soon as the columns penetrate a region where $\tau < \tau_y$, they are truncated: the magma solidifies there (Frigaard & Nouar, 2017; Balmforth & Craster, 2001). The band $|\phi| < 30^\circ$ is precisely the region where effective gravity is minimal (Maclaurin) and where the Coriolis stress is maximal ($\varepsilon \approx 55^\circ$): it is there, and only there, that $\tau > \tau_y$. Curved columns originating in the active band are thus naturally truncated at its boundaries.

Section synthesis

The CMT is a purely toroidal flow. The Taylor-Proudman objection does not apply. What remains to be demonstrated numerically — identified here as Limitation L1 (Section 22) — is whether a single torus forms within the band $|\phi| < 30^\circ$ rather than several alternating bands. This is a question for simulation, not a theoretical inconsistency.

7 Complete inventory of forces

All forces, projections, and equations are expressed in the instantaneous co-rotating Hadean frame of the body, with ε defined as in Section 3.

7.1 Complete force inventory: summary table

The CMT is a toroidal flow within a rapidly rotating fluid body, subject to a set of forces. Table 1 summarises all forces, their physical role, and their mathematical expression.

Table 1: Complete inventory of forces acting on the CMT.

Force	Role	Expression
Effective gravity	Radial confinement	$g_{\mathrm{eff}}(\phi) = g_{\mathrm{eff}}^{(0)}(1 + \frac{3}{2}J_2 \sin^2\phi) - \Omega^2 r(\phi)\cos^2\phi$
Centrifugal force	Lateral ejection, flattening	$F_{\mathrm{cent}} = \Omega^2 r(\phi)\cos^2\phi$
Radial Coriolis	Destabilisation, outward ejection	$F_{\mathrm{cor,rad}} = 2\Omega U \sin\varepsilon$
Horizontal Coriolis	Lateral confinement	$F_{\mathrm{cor,hor}} = 2\Omega U \cos\varepsilon$
Taylor-Proudman	Vertical columns, toroidal flow	$(\boldsymbol{\Omega}\cdot\nabla)\boldsymbol{u}_{\mathrm{pol}} = 0$; $\boldsymbol{u}_{\mathrm{tor}}$ unconstrained
Planetesimal bombardment	Stochastic perturbation, thermal reinjection	$\delta\boldsymbol{\Omega} = \sum_i \delta\boldsymbol{\Omega}_i$, $\delta U = \sum_i \delta U_i$
T-Tauri (CME)	Stochastic perturbation, thermal maintenance	$\boldsymbol{F}_{\mathrm{CME}}(t) = \sum_j \boldsymbol{F}_{\mathrm{CME},j}\,\delta(t - t_j)$
Solar tide	Low-frequency forcing on obliquity	$F_{\mathrm{tide,sol}} \sim 10^{-4} g_0 \cos(\omega_{\mathrm{sol}} t)$
Growing proto-lunar tide	Mathieu resonance (Episode 2+)	$F_{\mathrm{tide,moon}}(t) = A_m^{(k)}(t)\cos(n_{\mathrm{Moon}} t)$
Euler acceleration	Axial nutation	$\boldsymbol{a}_{\mathrm{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}$
Atmospheric pressure	Lateral confinement	$P_{\mathrm{atm}} \sim 10^5 - 10^7$ Pa
BH yield stress	Plasticity threshold	$\tau = \tau_y + k\dot{\gamma}^n$, $\tau > \tau_y$
Viscous stress	Dissipation	$\tau_{\mathrm{visc}} = \eta \nabla^2 \boldsymbol{u}$

7.2 Oscillation as a direct source of force

Axial oscillation implies $\dot{\boldsymbol{\Omega}} \neq \boldsymbol{0}$. Poincaré (1910) showed that this generates a Euler acceleration on every fluid parcel:

$$\boldsymbol{a}_{\mathrm{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}, \tag{11}$$

where $\boldsymbol{r}$ is the position vector. The radial projection:

$$\boldsymbol{a}_{\text{Euler}} \cdot \hat{r} = |\dot{\Omega}|\, r(\phi)\ \sin(\varepsilon + \phi), \tag{12}$$

is maximal within the band $|\phi| < 30°$ where $r(\phi)$ is largest: the oscillation is therefore not a mere ornament but a *direct energy source* for the CMT.

7.3 Radial Coriolis force and geometric threshold

The Coriolis force on a fluid parcel moving at azimuthal velocity U decomposes, in the oblique co-rotating frame, into:

$$f_{\text{rad}} = 2\,\Omega\ \sin\varepsilon \cdot U \quad \text{(radial, outward, destabilising)}, \tag{13}$$

$$f_{\text{horiz}} = 2\,\Omega\ \cos\varepsilon \cdot U \quad \text{(horizontal, confining)}, \tag{14}$$

with the ratio:

$$\frac{f_{\text{rad}}}{f_{\text{horiz}}} = \tan\varepsilon. \tag{15}$$

Geometric threshold $\varepsilon = 45°$ — necessary condition for radial ejection. For $\varepsilon > 45°$: $\tan\varepsilon > 1$, the destabilising radial component dominates the confining horizontal component. For $\varepsilon < 45°$: the horizontal component dominates and the Taylor-Proudman constraint inhibits radial ejection. This threshold follows purely from Coriolis kinematics: no adjustable parameter is involved. At $\varepsilon = 55°$: $f_{\text{rad}}/f_{\text{tot}} = \sin 55° \approx 0.82$, i.e. 82% of the total Coriolis force directed radially outward.

7.4 The T-Tauri Sun: catalyst and perturber

The Sun in its T-Tauri phase is not a driving force on the same footing as gravity or the Coriolis force — it does not act continuously on every CMT fluid parcel. It nonetheless plays three essential roles within the TPT window.

(1) Rheological modulator. T-Tauri UV/X-ray radiation keeps T_{surf} within the range $1,800$–$4,200$ K (Stefan-Boltzmann law, albedo $A_0 \approx 0.1$), which keeps the silicate magma's yield stress τ_y within the range 10^2–10^4 Pa (Giordano et al., 2008) — the optimal window for CMT cohesion. Below $T^* \approx 1,800$ K, τ_y grows to 5.5×10^3–8.0×10^4 Pa (Eq. 25) and cohesive ejection becomes progressively harder. The T-Tauri Sun therefore acts as a *guardian of the rheological window* via surface radiative forcing transmitted to the mantle by conduction through the silicate atmosphere.

(2) Source of stochastic perturbations. CMEs (10^{27}–10^{29} J per event, several per day) and radiative pulsations provide $\sim 10^9$–10^{10} opportunities to cross the potential barrier (Section 9.2.2) over 10^6 years. The number of opportunities is sufficient that the probability of no crossing at all is negligible. CMEs act as *stochastic triggers* for the first ejection episode, once dynamical conditions are met.

(3) Obliquity perturber. CMEs acting on the proto-atmosphere and magnetosphere deliver impulsive torques on the spin vector, contributing to the maintenance of the axial oscillation $\varepsilon \in [40°, 70°]$ (Section 3). The T-Tauri phase also provides an independent shutdown mechanism: once the Sun reaches the main sequence ($\approx$ 4.4 Ga), irradiation declines, τ_y increases, and the TPT window closes.

The T-Tauri Sun is therefore not the *engine* of the mechanism — gravity, the Coriolis force, the centrifugal force, Fe-Ni segregation, and Taylor-Proudman columns are — but it is the *catalyst* that keeps the window open and supplies the perturbations needed for stochastic crossings.

7.5 Tidal forces

Solar tide. Relative amplitude $\approx 10^{-4} g_0$; contributes as a low-frequency resonant forcing on obliquity.

Growing proto-lunar tide — Mathieu resonance. Starting from Episode 2, the proto-satellite exerts a tidal force that modulates the height of the potential barrier:

$$\Delta E_{\text{barr}}^{(k)}(t) = \Delta E_{\text{barr}}^{(k,0)} - A_m^{(k)}(t) \cos(n_{\text{Moon}}\, t), \tag{16}$$

where $A_m^{(k)}(t)$ is the growing tidal amplitude and n_{Moon} the proto-lunar mean motion. This gives the radial perturbation equation:

$$\ddot{\xi}_r = \left[\lambda + A_m^{(k)}(t) \cos(n_{\text{Moon}}\, t)\right] \xi_r, \tag{17}$$

which is the Mathieu equation (McLachlan, 1947). Parametric resonance bands exist for $n_{\text{Moon}} \approx 2\sqrt{|\lambda|}$ even when $\lambda < 0$: the growing proto-lunar tide is the dominant trigger for Episodes 2 through N.

Giant-planet tides. Mainly Jupiter: a low-frequency perturbation on obliquity, sustaining the oscillation over timescales of 10^5–10^6 yr.

7.6 The parametric elliptical instability as a CMT amplifier

In a precessing fluid ellipsoid, precessional flows can resonate with the fluid's inertial modes (Poincaré waves, frequencies $\omega_{\text{in}} = 2\Omega m/n$). This mechanism — the parametric elliptical instability — was established theoretically by Malkus (1968), formalised by Kerswell (2002), and demonstrated experimentally by Lacaze et al. (2004).

For the Maclaurin spheroid geometry at $T_{\text{rot}} = 3.5$ h and $f \approx 0.075$, the free nutation frequency is $\omega_E = \Omega f \approx 3.74 \times 10^{-5}$ rad s^{-1} (Eq. 7). This frequency is commensurate with the fluid's inertial modes by geometric construction — not by parameter tuning.

The linear growth rate is:

$$\sigma_{\text{res}} = \frac{\Omega f}{2} \approx 1.87 \times 10^{-5}\,\text{rad}\,\text{s}^{-1}. \tag{18}$$

The CMT velocity grows exponentially:

$$U_{\text{CMT}}(t) = U_{\text{turb}}\, e^{\sigma_{\text{res}} t}, \qquad U_{\text{turb}} \approx 0.8\,\text{km}\,\text{s}^{-1}, \tag{19}$$

where U_{turb} is the characteristic velocity at the Rhines scale (Rhines, 1975). The time to reach $U_{\text{crit}} \approx 9.8$ km s^{-1} is:

$$t_{\text{acc}} = \frac{\ln(U_{\text{crit}}/U_{\text{turb}})}{\sigma_{\text{res}}} \approx 37\,\text{h}, \tag{20}$$

with $\ln(U_{\text{crit}}/U_{\text{turb}}) = \ln(9.8/0.8) \approx 2.51$.

The CMT reaches bifurcation velocity within $\approx$ 37 hours. This result is entirely determined by $f \approx 0.075$ and $\Omega = 4.99 \times 10^{-4}$ rad s^{-1}, imposed by $T_{\text{rot}} = 3.5$ h and Maclaurin-spheroid physics. The elliptical instability thus acts as a powerful amplifier, carrying the CMT to bifurcation velocity within $\approx$ 37 hours — a timescale fully consistent with the TPT chronology.

As Fe-Ni segregation proceeds, CMT density decreases:

$$\rho_{\text{CMT}}(t) = \rho_0 - \xi_{\text{Fe}}(t)(\rho_0 - \rho_f), \quad \rho_0 \approx 4{,}500\,\text{kg}\,\text{m}^{-3},\ \rho_f \approx 3{,}200\,\text{kg}\,\text{m}^{-3}, \tag{21}$$

which amplifies all destabilising terms via the factor $\rho_0/\rho_{\text{CMT}}(t)$ in the instability equation (Section 9.2).

7.7 Geostrophic β force and the Rhines scale

The β parameter, measuring the latitudinal gradient of the Coriolis parameter:

$$\beta = \frac{2\Omega\cos\varepsilon}{R_{\text{eq,proto}}} \approx 8.98 \times 10^{-11}\,\text{m}^{-1}\,\text{s}^{-1} \quad (\varepsilon = 55^\circ), \tag{22}$$

arrests the inverse cascade at the Rhines scale (Rhines, 1975):

$$L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}} \approx 3.34 \times 10^{6}\,\text{m} \approx 0.52\,R_\oplus, \tag{23}$$

where U_{turb} is the turbulent velocity scale. This scale $L_\beta \approx 0.52\,R_\oplus$ corresponds to harmonic mode $\ell = 1$, $m = 1$: the largest coherent structure possible on a sphere, and the spatial expression of the CMT.

8 Hierarchy of hypotheses

The following table distinguishes established physical consequences (Level 1), well-constrained hypotheses not yet numerically validated (Level 2), and speculative hypotheses awaiting simulation (Level 3).

Table 2: Explicit hierarchy of hypotheses underlying the TPT.

ID	Hypothesis / Statement	Level	Depends on
H0	Central thesis: the Moon formed through three coupled transitions triggered by Fe-Ni segregation	2 (constrained)	Thermodynamic bound $E_{\mathrm{grav}}/E_{\mathrm{fus}} \approx 155$
H1	Total volumetric melting: the entire silicate mantle above the liquidus during the 4.568–4.4 Ga window	1 (established)	Factor $E_{\mathrm{grav}}/E_{\mathrm{fus}} \approx 155$ (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015)
H2	$\varepsilon \in [40°, 70°]$: sustained free nutation in the body's co-rotating frame	2 (constrained)	Absence of Moon + 5 convergent arguments (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014)

ID	Hypothesis / Statement	Level	Depends on
H3	CMT confinement to $\|\phi\| < 30°$ admits an effective description as the ground-state mode of a nonlinear Schrödinger-type equation with effective potential $V_{\text{eff}}(\phi)$; the Gaussian solution is the first-order (linearised) approximation of this mode	3 (speculative)	Reduction from the potential vorticity equation (Redekopp, 1977; Luo, 1991; Yin et al., 2019); conditions $\tau_y \lesssim 10^2$ Pa, $\varepsilon > 45°$; quantitative validity of the Gaussian approximation not yet established (Limitation L12)
H4	The $\ell = 1$ mode is selected by the combination of BH rheology (suppression of $\ell < \ell_{\text{min}}$ modes) and the inverse cascade	3 (speculative)	3D SPH simulation with full BH rheology (L1)
H5	$\tau_y(T)$ follows an Arrhenius law with $T^* \approx 1,800$ K and $\tau_y \lesssim 10^2$ Pa for $T \gtrsim 4,000$ K	1 (established)	Laboratory rheology (Giordano et al., 2008)
H6	The double-well potential $V(U)$ emerges from coupling between Fe-Ni segregation and angular-momentum loss via ejection	2 (constrained)	Coupled system (Eqs. 28, 29); P22
H7	The Kramers crossing rate applies with $N_{\text{opp}} \sim 10^9$–10^{10} stochastic opportunities	2 (constrained)	Statistical physics (Kramers, 1940) + CMEs + impacts

ID	Hypothesis / Statement	Level	Depends on
H8	$f_{\text{cap}} \approx 0.70$ and $\partial f_{\text{cap}}/\partial M_{\text{Moon}} > 0$	3 (speculative)	3D SPH simulation (L2); lower bound $f_{\text{cap}}^{\min} \gtrsim 0.45$ (Appendix G)
H9	Dynamo delay = 350 ± 50 Myr, sum of three independent phases	2 (constrained)	Hf-W (Kleine et al., 2002; Yin et al., 2002) + zircons (Wilde et al., 2001; Valley et al., 2002) + Tarduno et al. (2025)
H10	Seismic preservation window: 200–530 km	2 (constrained)	Ilmenite cumulate overturn (Briaud et al., 2023) + mare volcanism (Li et al., 2021; Che et al., 2021)
H11	Two-phase *'aā*-type CMT structure: mobile plastic core + fragmented clinker crust	3 (speculative)	3D SPH simulation (L1, L3)
H12	$\mathrm{Bi}_{\mathrm{KH}} \gg 1$ suppresses Kelvin-Helmholtz fragmentation at large and intermediate scales	2 (constrained)	Frigaard & Nouar (2017); three independent confinement mechanisms

ID	Hypothesis / Statement	Level	Depends on
H13	$b' = 1$ is a dynamical attractor of the coupled segregation–ejection system	2 (constrained)	Geostrophic equilibrium + angular-momentum conservation (Appendix B)
H14	The inverse cascade is arrested at the Rhines scale $L_\beta \approx 0.52\,R_\oplus$; the peak at $\ell \approx 14$ is redistributed toward $\ell = 1$ by nonlinearity and BH rheology	3 (speculative)	3D SPH simulation (L1)
H15	The ejected flow transits through three phases: ballistic expansion $\rightarrow$ orbital insertion $\rightarrow$ cohesive accretion beyond R_{Roche}	3 (speculative)	3D SPH simulation (L6)
H16	The parametric elliptical instability amplifies the CMT with growth rate $\sigma_{\text{res}} = \Omega f/2$	2 (constrained)	Malkus (1968); Kerswell (2002); Lacaze et al. (2004); Maclaurin geometry
H17	The T-Tauri phase maintains $\tau_y \lesssim 10^2$ Pa through irradiation, enabling the $\ell = 1$ mode	2 (constrained)	Stefan-Boltzmann law + BH rheology; testable by simulation

9 The three coupled transitions

The single engine is the progressive segregation of Fe-Ni toward the forming core. It simultaneously governs three regime changes: the rheological transition opens the window, the mechanical transition produces the ejection episodes, and the magnetic transition closes the loop with the terrestrial dynamo.

9.1 Transition 1 — Rheological: the Bingham-Herschel law

A Newtonian fluid flows under any applied stress. A Bingham-Herschel (BH) fluid, by contrast, resists like a solid as long as the applied stress does not exceed a plasticity threshold τ_y (Bingham, 1922; Herschel & Bulkley, 1926):

$$\tau = \tau_y + k\dot{\gamma}^n \quad (\tau > \tau_y), \tag{24}$$

where τ is the applied shear stress, τ_y the yield stress, k the consistency index, $\dot{\gamma}$ the shear rate, and n the flow index. For a partially crystallised silicate magma at a solid fraction $\phi \approx 0.1$–0.3: $n = 1.2$ (inverse shear-thinning, or shear-thickening, Caricchi et al. 2007), $\tau_y \in [10^2, 10^4]$ Pa at $T = 3,000$–$3,500$ K (Giordano et al., 2008).

The yield stress depends exponentially on temperature (Arrhenius law):

$$\tau_y(T) = \tau_y^{(0)} \exp\left(+\frac{E_a}{RT}\right), \qquad E_a \approx 150\text{–}250\,\mathrm{kJ\,mol^{-1}}, \tag{25}$$

where R is the gas constant and E_a the activation energy — the positive sign in the exponential reflects the fact that τ_y *increases* as T *decreases*, as for the viscosity of a liquid (Giordano et al., 2008). At $T = 3,000$ K: $\tau_y \approx 10^2$ Pa — the flow moves easily. At $T^* \approx 1,800$ K, depending on the value adopted for E_a within the 150–250 $\mathrm{kJ\,mol^{-1}}$ range, τ_y grows to 5.5×10^3–8.0×10^4 Pa — an increase of one to three orders of magnitude relative to $T = 3,000$ K, sufficient that cohesive ejection becomes progressively harder as the T-Tauri Sun leaves its active phase and the surface cools.

This is the natural thermal shutdown mechanism at $T^* \approx 1,800$ K: the TPT window closes irreversibly by the Arrhenius law alone, with no additional assumption. This is not a fitted parameter: it is a thermodynamic consequence of the exponential temperature dependence of τ_y.

The CMT's 'aā structure

The toroidal flow displays a two-phase architecture: a mobile, high-temperature plastic core surrounded by a fragmented clinker crust. This is not a cohesion problem: it is its guarantee. The outer crust protects the core against Kelvin-Helmholtz instabilities at the flow/atmosphere interface. A quenched glassy shell (analogous to the glassy margins of pillow lavas, Hekinian et al. 1989), forming in ≈ 7 minutes upon contact with the atmosphere, completes this rheological shield. This structure is consistent with the work of Frigaard & Nouar (2017) and Balmforth & Craster (2001).

9.2 Transition 2 — Mechanical: instability and ejection

9.2.1 *The generalised instability equation*

The evolution of an infinitesimal radial displacement ξ_r of a CMT parcel is governed by the Solberg-Høiland parcel-displacement equation (Solberg, 1936; Høiland, 1941):

$$\ddot{\xi}_r = \lambda\,\xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement).} \tag{26}$$

Let $a = -(1+\alpha)/r^2$, where α is the rotation-profile exponent ($U \propto r^\alpha$). Let also $b = 2\Omega^{(k)}\sin\varepsilon > 0$ and $c = g_{\text{eff}}^{(k)} < 0$. Dimensional consistency (Appendix C) requires a $1/r$ factor on the gravity and Coriolis terms, absent from earlier versions of this work. The stability coefficient is then:

The generalised instability coefficient

$$\lambda(U,\varepsilon,\alpha,k) = \underbrace{\frac{c}{r}}_{\substack{\lambda_1<0\\ \text{gravity}}} + \underbrace{\frac{b}{r}U}_{\substack{\lambda_2>0\\ \text{rad. Coriolis}}} + \underbrace{aU^2}_{\substack{\lambda_3<0\\ \text{ang. mom.}}} + \underbrace{\frac{|\dot{\Omega}|r(\phi)\sin(\varepsilon+\phi)}{R_{\text{eq,proto}}}}_{\substack{\lambda_4\\ \text{Euler/oscillation}}} + \underbrace{\frac{g_{\text{eff}}^{(0)}J_2\sin(2\phi)}{R_{\text{eq,proto}}}}_{\substack{\lambda_5\\ \text{Maclaurin grad.}}} \tag{27}$$

Each term has units of s^{-2} (Appendix C) and a precise physical meaning. $\lambda_1 < 0$ is the stabilising effective gravity. $\lambda_2 > 0$ is the destabilising radial Coriolis force; it grows with ε. $\lambda_3 < 0$ is the angular-momentum gradient term; it is always stabilising for any physical profile ($\alpha > -1$), consistent with the classical Rayleigh stability criterion, and its magnitude ($\sim 10^{-7}$) is negligible compared with the dominant terms. λ_4 is the Euler acceleration due to axial oscillation. λ_5 is the tangential gravity gradient of the Maclaurin geometry.

The amplification factor $\rho_0/\rho_{\text{CMT}}(t)$ (Eq. 21) monotonically amplifies all destabilising terms, progressively pushing λ toward zero.

9.2.2 *The double-well potential: emergence of the coupled system*

Integrating $\lambda(U) = -dV/dU$ gives an effective potential $V(U)$ of cubic form (Appendix C). Whether this potential exhibits a double-well structure — a confined well P_1 where the CMT accumulates energy, and an ejective well P_2 where cohesive ejection becomes favourable, separated by a barrier $\Delta E_{\text{barr}}^{(k)}$ — depends on the sign of the discriminant Δ of the underlying quadratic equation (Appendix C, Eq. 66). Evaluated at the nominal initial parameters (Appendix C, Table 7), this condition is *not* satisfied: the proto-Earth does not begin in a double-well configuration.

Evolution of CMT density. Fe-Ni segregation is the TPT's single engine. The segregated Fe-Ni fraction, $\xi_{\text{Fe}}(t)$, evolves according to:

$$\frac{d\xi_{\text{Fe}}}{dt} = \frac{1}{\tau_{\text{seg}}}\left[\xi_{\text{Fe}}^{\text{sat}} - \xi_{\text{Fe}}(t)\right] \tag{28}$$

where $\xi_{\text{Fe}}^{\text{sat}} \approx 0.15$ is the saturated Fe-Ni fraction in the mantle. CMT density evolves as $\rho_{\text{CMT}}(t) = \rho_0[1 - \xi_{\text{Fe}}(t)]$, and, by Gauss's law for a spherical shell, $|g_{\text{eff}}|(t) = |g_{\text{eff}}|_0[1 - \xi_{\text{Fe}}(t)]$.

Evolution of angular velocity. Angular velocity $\Omega(t)$ evolves under two opposing processes: ejection loss, where each ejection carries off a fraction $\epsilon = M_{\text{ej}}/M_{\text{CMT}}$ of the CMT's angular momentum, and contraction acceleration, where cooling and differentiation contract the proto-Earth, increasing Ω through the skater effect. The evolution equation is:

$$\frac{d\Omega}{dt} = -\frac{\Omega}{\tau_{\text{ej}}} + \frac{\Omega}{\tau_{\text{contr}}} \tag{29}$$

where τ_{ej} is the timescale between ejections and τ_{contr} the contraction timescale.

The coupled system. Equations (28) and (29) form a coupled system. The coupling operates via Δ: ξ_{Fe} decreases $|g_{\text{eff}}|$, which increases Δ; Ω decreases b, which decreases Δ. The double well emerges when $\Delta > 0$. The solution of Eq. (28) is $\xi_{\text{Fe}}(t) = \xi_{\text{Fe}}^{\text{sat}}[1 - e^{-t/\tau_{\text{seg}}}]$, and the solution of Eq. (29) is $\Omega(t) = \Omega_0 \exp[-t/\tau_{\text{ej}} + t/\tau_{\text{contr}}]$. During the active TPT window (30–50 Myr), ejections dominate over contraction: $1/\tau_{\text{ej}} > 1/\tau_{\text{contr}}$, so that $\Omega(t)$ decreases.

The emergence threshold. The criterion $\Delta(t) > 0$ is a condition on two variables: $\Omega(t)$ (which decreases through angular-momentum loss) and $|g_{\text{eff}}|(t)$ (which decreases through Fe-Ni segregation). These two evolutions are coupled through Eqs. (28) and (29). Evaluating the criterion in two limiting cases, at the nominal parameters of Table 7: with $\Omega = \Omega_0$ fixed, the threshold is reached once $|g_{\text{eff}}|$ has decreased by about **91%** relative to its initial value; with $|g_{\text{eff}}| = |g_{\text{eff}}|_0$ fixed, the threshold cannot be reached by a decrease in Ω — it would instead require an *increase* in Ω by a factor of ≈ 3.4. This second limiting case is therefore physically out of reach for the system as described (where Ω only decreases during the active window), and only the first — the reduction of $|g_{\text{eff}}|$ via Fe-Ni segregation, combined with the complementary mechanisms of Appendix H — can effectively trigger the double well at the nominal parameters adopted here.

P22 — The unified criterion for double-well emergence

At the proto-Earth's nominal global parameters, the criterion $\Delta > 0$ requires a reduction of $|g_{\text{eff}}|$ of about 91% at fixed Ω (Appendix C, Section C.4). The complementary mechanisms of Appendix H — reduced effective CMT density via extended segregation and vesicularity, obliquity at its geometric maximum, and angular-momentum input from planetesimal bombardment — close this gap and render the criterion $\Delta > 0$ attainable within the active TPT window. The falsifiable prediction concerns crossing this complete critical surface, testable by 3D SPH simulation following the coupled evolution of the system. The complete budget is given in Appendix H.

Falsification: the absence of this threshold in 3D SPH simulations coupling CMT dynamics to Fe-Ni segregation (Limitations L1, L2), or a threshold value markedly different from projected values, would weaken the current interpretation of ejection episodes as Kramers-type barrier crossings, without on its own ruling out the broader TPT framework if an alternative ejection mechanism could be identified.

The episode sequence. Once the double well has emerged, the Kramers crossing rate (Kramers, 1940) applies:

$$\Gamma_{\text{Kramers}} = \frac{\omega_0 \omega_b}{2\pi\gamma_{\text{diss}}} \exp\left(-\frac{\Delta E_{\text{barr}}}{k_B T_{\text{eff}}}\right), \tag{30}$$

where ω_0 and ω_b are the oscillation frequencies at the bottom of the well and the top of the barrier respectively, γ_{diss} the dissipation rate, and $k_B T_{\text{eff}}$ the effective thermal energy available to the CMT. Crossing the barrier triggers an ejection, which decreases Ω and lowers the barrier height ΔE_{barr}, making the next crossing more probable. The process repeats until ejectable material is exhausted: the number $N = 2$–3 emerges from the coupled system as a consequence of the progressive decay of the barrier, not as an a priori assumption.

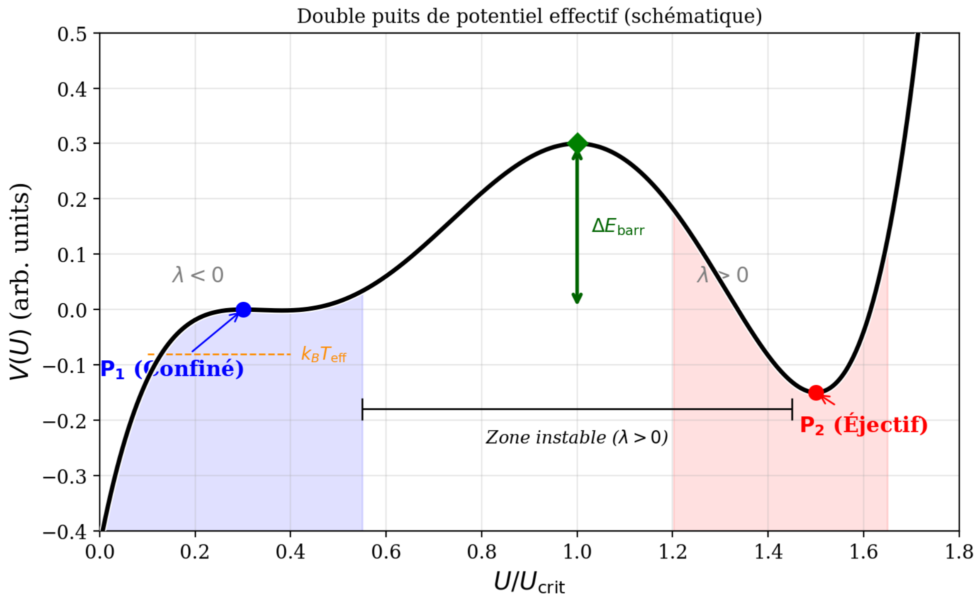

Figure 1: Schematic representation of the effective double-well potential $V(U)$. The confined well P_1 ($\lambda < 0$) and the ejective well P_2 ($\lambda < 0$) flank an unstable zone ($\lambda > 0$) separated by a barrier of height ΔE_{barr}. Thermally activated crossing, governed by the Kramers rate (Eq. 30), is rendered near-certain by the available effective thermal energy, $k_B T_{\text{eff}}$, and the large number of stochastic opportunities.

9.2.3 *Critical velocity and maximum ejectable mass*

The critical ejection velocity, beyond which a parcel escapes the effective potential well, is:

$$U_{\text{crit}}^{(0)} = \sqrt{2\, g_{\text{eff}}^{(0)}\, R_{\text{eq,proto}}} \approx 9.8\,\text{km}\,\text{s}^{-1}, \tag{31}$$

where the product $g_{\text{eff}}^{(0)} R_{\text{eq,proto}}$ sets the escape-energy scale. The ejection Mach number, using the sound speed in two-phase magma $c_s \approx 0.9\ \text{km}\,\text{s}^{-1}$ (Mader et al., 2013), is $\text{Ma} = U_{\text{crit}}/c_s \approx 10.9$: the ejection is **hypersonic**.

The maximum ejectable mass per episode is a fraction ϵ of the CMT mass,

$$M_{\text{ej}}^{\text{max}} \approx \epsilon\, M_{\text{CMT}} \approx 2.0\text{–}4.0 \times 10^{22}\,\text{kg}, \tag{32}$$

where $\epsilon \approx 0.1$ (Appendix B) is the mass fraction carried off by Kramers barrier crossing, and M_{CMT} the CMT mass. Ejectable mass increases with faster rotation (large $\Omega^{(k)}$), more favourable obliquity ($\sin\varepsilon$ close to 1), larger CMT mass, and weaker confining gravity (g_{eff} reduced by Fe-Ni depletion and angular-momentum loss).

At $T_{\text{rot}} = 3.5$ h, with $M_{\text{CMT}} \approx 3.3 \times 10^{23}$ kg and $\epsilon \approx 0.1$: $M_{\text{ej}}^{\text{max}} \approx 3.3 \times 10^{22}$ kg $\approx 0.45\, M_{\text{Moon}}$, consistent with the 2.0–4.0 $\times 10^{22}$ kg range adopted in the mass budget (Section 10). This is why $N = 2$–3 episodes suffice.

9.2.4 *Cohesion of the ejected flow: the* $\mathrm{Bi_{KH}}$ *criterion*

For free-streaming flow, the Weber number $We \approx 1.4 \times 10^{15}$ would imply fragmentation by atomisation. However, the CMT is not a free-streaming flow: it is a collective, self-gravitating toroidal flow, confined by toroidal symmetry, self-gravitation, and atmospheric pressure.

The correct criterion is the Bingham-Kelvin-Helmholtz number, which compares the yield stress to the Kelvin-Helmholtz shear stress: for $\mathrm{Bi_{KH}} \gg 1$, the yield stress suppresses the KH instability and cohesion is maintained (Frigaard & Nouar, 2017). Three independent conditions reinforce cohesion: (1) atmospheric confinement ($P \approx 10^5$–10^7 Pa); (2) internal crystal network ($\phi \approx 0.1$–0.3, multiplying fragmentation resistance by a factor of 3 to 10); (3) density ratio $\rho_{\mathrm{flow}}/\rho_{\mathrm{atm}} \approx 200$ (Rayleigh-Taylor suppression). The complete demonstration of the $\mathrm{Bi_{KH}}$ criterion is given in Appendix D.

9.2.5 *Transient post-ejection precession and crustal dichotomy*

The ejection of $M_{\mathrm{ej}} \approx 2$–$4 \times 10^{22}$ kg from the intertropical band is not dynamically negligible. Asymmetric mass loss alters the inertia tensor, producing: (1) a decrease $\Delta\Omega/\Omega \approx -10$–$15\%$, lowering g_{eff} and U_{crit} for the next episode; (2) shape oscillations as the spheroid contracts toward a rounder sphere; (3) a reorganisation of obliquity toward a new value within $[40°, 70°]$.

This last point has a direct geometric consequence, not merely a numerical one: since the intertropical band $|\phi| < 30°$ is defined perpendicular to the instantaneous axis $\mathbf{\Omega}(t)$ (Section 3), a new value of ε shifts the position of the dynamical equator $\phi = 0$ itself, and hence the entire active band, relative to the body. The intertropical band hosting the second episode is therefore not, in general, the same region of the body that hosted the first: same angular width ($\pm 30°$), same definition relative to $\mathbf{\Omega}(t)$, but a different location on the spheroid. It is this geometric displacement of the active band, episode after episode, that directly causes the crustal asymmetry described below, and not merely the change in the numerical value of ε.

This displacement of the active band, combined with the shape oscillations of point (2) above, has a further consequence for the CMT itself: the torus that re-forms for the next episode does not reproduce an identical, merely displaced copy of the previous one. It forms on a spheroid whose own shape has slightly changed, along a band whose position has itself changed. The next CMT is therefore ejected toroidally in a manner distinct from the previous one, both in shape and orientation. This remains, at this stage, a qualitative statement: no quantitative derivation of the shape or angular offset between successive episodes is attempted here.

The second episode ejects a cohesive toroidal flow onto a still hot and deformable proto-POB, within a limited solid angle (corresponding to the projected intertropical band), which becomes the present-day farside. Three mechanisms amplify the resulting crustal asymmetry: early gravitational locking ($\tau_{\text{lock}} \sim 10^4$–$10^5$ yr (Wieczorek et al., 2013)), the residual obliquity of the proto-POB, and irreversible BH plastic welding. The $\approx 2:1$ thickness ratio (nearside $\approx$ 34 km versus farside $\approx$ 54 km, Wieczorek et al. 2013) is a qualitative prediction of the TPT.

9.3 Transition 3 — Magnetic: the progressive dynamo

The $\approx$ 350 Myr delay between the inferred time of lunar formation ($\approx$ 4.55 Ga) and the first geological detection of Earth's magnetic field ($\approx$ 4.2 Ga, Tarduno et al. 2025) is not a fitted parameter in the TPT. It is proposed as the sum of three independently constrained durations:

$$\Delta t_{\text{dynamo}} = \underbrace{\Delta t_{\text{ej}}}_{\approx 60\,\text{Myr}} + \underbrace{\Delta t_{\text{cool}}}_{\approx 140\,\text{Myr}} + \underbrace{\Delta t_{\text{emer}}}_{\approx 150\,\text{Myr}} \approx 350\,\text{Myr}. \tag{33}$$

Phase 1 ($\approx$ 60 Myr) — Active Fe-Ni segregation. Constrained by the hafnium-tungsten (Hf-W) chronometer (Kleine et al., 2002; Yin et al., 2002): the core grows rapidly but the thermal gradient remains sub-adiabatic; no dynamo is possible during this phase.

Phase 2 ($\approx$ 140 Myr) — Protocrust formation. The growing Moon stabilises obliquity at $\approx 23.5°$; the protocrust forms around 4.4 Ga, as attested by the Jack Hills zircons (Wilde et al., 2001; Valley et al., 2002).

Phase 3 ($\approx$ 150 Myr) — Progressive dynamo emergence. Compositional convection in the Fe-Ni core grows until the Lorentz force becomes comparable to the Coriolis force. The Elsasser number Λ balances these two forces. The threshold $B_{\text{crit}} \approx 2.6$ mT is derived in Appendix E. Modern geodynamo simulations (Christensen & Aubert, 2006; Olson & Christensen, 2006) show that the transition is progressive, not abrupt: $\Lambda \sim 1$ is an order-of-magnitude benchmark, not a strict threshold.

Section synthesis

The $\approx$ 350 Myr delay involves no free parameter: Δt_{ej} is constrained by the Hf-W chronometer (Kleine et al., 2002; Yin et al., 2002); Δt_{cool} is constrained by the Jack Hills zircons at 4.4 Ga (Wilde et al., 2001; Valley et al., 2002); and Δt_{emer} is constrained by Tarduno et al. (2025) at 4.2 Ga. These are three independent observations, not an interpolation. The associated uncertainty is $\pm$50 Myr (Limitation L4, Section 22).

10 Mass budget: $N = 2$–3 episodes

10.1 Corrected target lunar mass

A fundamental point: the currently observed lunar mass $M_{\text{Moon}} = 7.34 \times 10^{22}$ kg is *not* the mass that the TPT mechanism must produce. The Moon accreted additional mass after the ejection episodes: a differentiated body about 260 km in diameter struck the Moon at the South Pole–Aitken basin and *remained there*, contributing $M_{\text{SPA}} \approx 3.2 \times 10^{19}$ kg (Wakita et al., 2026) — this material is now part of the observed lunar mass — and a documented late accretion contributed an additional mass ΔM_{late} (Zellner, 2019). The target mass for the TPT mechanism is therefore:

$$M_{\text{TPT}} = M_{\text{Moon}} - M_{\text{SPA}} - \Delta M_{\text{late}} < 7.34 \times 10^{22} \text{ kg}. \tag{34}$$

10.2 Mass-budget equation

The mass-budget equation

$$M_{\text{Moon}} = N \cdot f_{\text{cap}} \cdot M_{\text{ej}}^{\text{max}} + M_{\text{SPA}} + \Delta M_{\text{late}}, \tag{35}$$

with $N = 2$–3 (nominal), $f_{\text{cap}} \in [0.65, 0.85]$ (nominal 0.70), $M_{\text{ej}}^{\text{max}} \approx 2.0$–$4.0 \times 10^{22}$ kg per episode.

f_{cap}	$M_{\text{TPT}}^{(N=2)}$	$M_{\text{TPT}}^{(N=3)}$	Fraction of M_{Moon}
0.50	2.5×10^{22} kg	3.7×10^{22} kg	34–50%
0.70	3.5×10^{22} kg	5.2×10^{22} kg	48–71%
0.85	4.3×10^{22} kg	6.4×10^{22} kg	59–87%

The remainder is explained by M_{SPA} and documented late accretion. The budget is robust across the entire f_{cap} range.

f_{cap} is an increasing state variable: $\partial f_{\text{cap}} / \partial M_{\text{Moon}} > 0$, since the larger the proto-Moon, the more efficiently it captures subsequent ejecta. This positive feedback, combined with the decrease of $g_{\text{eff}}^{(k)}$ at each episode, defines a dynamical attractor that reinforces the mechanism.

10.3 f_{cap} as a testable prediction, not a free parameter

The position adopted here is that f_{cap} should not be treated as an adjustable parameter whose wide range would render the mass budget uninformative, but as a quantity that the TPT predicts quantitatively and which can, in principle, falsify the mechanism.

Physical basis: the classical disc-accretion range. Reference N-body simulations of lunar accretion from an impact-generated debris disc (Kokubo et al., 2000) establish that the efficiency of incorporating disc material into a final moon is 10 to 55%, this efficiency increasing with the disc's initial specific angular momentum. This result, robust across a wide range of initial conditions and numerical resolutions, constitutes the external benchmark against which f_{cap} must be compared.

The gap as a potential signature. The nominal value $f_{\mathrm{cap}} \approx 0.70$ adopted in the TPT is markedly above the upper bound of this classical range (55%). Within the TPT framework, this gap is not accidental: it is attributed to the cohesion of the ejected toroidal flow and its orbital concentration within a limited solid angle (Appendix G), unlike a debris disc dispersed over 4π steradians as in the classical giant-impact scenario. If this interpretation is correct, the gap between the TPT's adopted value of f_{cap} and the classical range becomes a discriminating test rather than mere parametric latitude.

Falsifiable Prediction — P17 — Capture efficiency beyond the classical disc-accretion range

The orbital coherence of the TPT's ejected toroidal flow carries effective capture efficiency beyond the classical 10–55% range established for accretion from a dispersed debris disc (Kokubo et al., 2000). The TPT predicts $f_{\mathrm{cap}} \gtrsim 0.65$–$0.70$.

Falsification: if a value of f_{cap} derived from a dedicated 3D SPH simulation of the CMT geometry (BH rheology, atmospheric confinement, orbital concentration within the intertropical band) falls within the classical 10–55% range rather than beyond it, this aspect of the theory is falsified. A measured f_{cap} within the classical range would not necessarily invalidate the TPT as a whole (the mass budget would remain attainable via M_{SPA} and late accretion), but it would strip f_{cap} of its status as a distinctive signature relative to the giant-impact model.

Acknowledged limitation: this prediction rests on the assumption that orbital concentration of the TPT flow (Appendix G) produces an accretion regime qualitatively different from the classical dispersed disc. This assumption is itself Level 3 (not quantitatively demonstrated), so that this becomes a robust test only once the 3D SPH simulation is available. Until then, it remains a qualified prediction: the gap with the classical range is consistent with the TPT, but does not independently confirm it.

11 Angular-momentum transport by ejecta

Within the present framework, angular-momentum loss is not treated as a single-velocity ballistic process, but as a distributed transport mechanism resulting from a spectrum of ejected material generated over successive differentiation-driven phases.

11.1 Ejecta distribution function

We describe the mass distribution of ejecta as a function of velocity and emission angle:

$$dM = \rho(v, \theta)\, dv\, d\theta, \tag{36}$$

where v is the ejection velocity, θ is the angle relative to the local tangential direction, and $\rho(v, \theta)$ is the phase-dependent ejecta distribution function.

The distribution $\rho(v, \theta)$ is not arbitrary. It follows from three physical processes: the CMT's velocity spectrum is not monokinetic, turbulent fluctuations at the Rhines scale (Section 7.6) producing a velocity spread $\delta v_{\rm CMT} \sim U_{\rm turb} \approx 0.8\ {\rm km\,s^{-1}}$; the double-well potential (Section 9.2.2) means barrier crossing occurs over a range of energies, and the Kramers rate (Eq. 30) implies a Gaussian-type distribution $\rho_v(v) \propto \exp[-(v-\bar{v})^2/2\sigma_v^2]$ with $\bar{v} \approx U_{\rm crit}$ and $\sigma_v \sim (k_B T_{\rm eff}/M_{\rm CMT})^{1/2}$; and the finite extent of the torus produces an angular spread $\rho_\theta(\theta) \propto \cos\theta$ for $|\theta| < \theta_{\rm max}$, with $\theta_{\rm max} \approx 30^\circ$.

For analytical tractability, we adopt the separable form:

$$\rho(v, \theta) = M_{\rm ej}\, \mathcal{N}_v\, \mathcal{N}_\theta\, \exp\left[-\frac{(v-\bar{v})^2}{2\sigma_v^2}\right] \cos\theta, \quad v > v_{\rm esc},\ |\theta| < \theta_{\rm max}, \tag{37}$$

where $v_{\rm esc} = \sqrt{2 g_{\rm eff} R_p}$. This distribution is motivated by the underlying physics, not freely chosen: it is built from the CMT's turbulent velocity spread, energy-dependent Kramers crossing (Appendix C), and the finite angular extent of the torus. No free parameter is introduced beyond those already present in the theory developed in the preceding sections.

11.2 Specific angular momentum

The specific angular momentum of an ejecta element is $h = r\, v \sin\theta$, where r is the effective emission radius. This radius is not simply the planetary radius R_p: the silicate atmosphere (Section 2) extends the effective surface to $r_{\rm atm} \approx R_p + H$, where the scale height $H \approx 200$ km for $T \approx 3000$ K and $g \approx 6\ {\rm m/s^2}$; and the toroidal flow is raised above the surface by centrifugal effects, the CMT's centre lying at $r_{\rm CMT} \approx R_p + U_{\rm CMT}^2/(2g_{\rm eff}) \approx R_p + 500$ km. We therefore adopt $r \approx R_p + \Delta r$, with $\Delta r \approx 300$–500 km. This correction introduces only an uncertainty of a few percent in the angular-momentum budget, well within the overall error margin.

11.3 Angular-momentum loss

The total angular momentum carried off by non-accreted material is:

$$L_{\rm loss} = \int\!\!\int \left(1 - f_{\rm cap}(v, \theta)\right)\, r\, v \sin\theta\, \rho(v, \theta)\, dv\, d\theta. \tag{38}$$

Capture efficiency $f_{\text{cap}}(v,\theta)$ is not a constant: it depends on the orbital fate of each ejecta element,

$$f_{\text{cap}}(v,\theta) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{v\sin\theta - v_{\text{crit}}}{\sigma_f}\right)\right], \tag{39}$$

where $v_{\text{crit}} \approx v_{\text{circ}} = \sqrt{g_{\text{eff}} R_p}$ is the circular orbital velocity and σ_f a smoothing width representing the finite transition between captured and lost material. Rather than treating f_{cap} as a single ad hoc value, we express it here as a function $f_{\text{cap}}(v,\theta)$ of orbital mechanics. The nominal value $f_{\text{cap}} \approx 0.70$, independently estimated from the CMT's cohesive fraction and the re-accretion of co-orbital fragments, is consistent with the mass-weighted average of this function over the proposed ejecta spectrum.

11.4 Discrete formulation and Monte Carlo implementation

For a numerical implementation, the expression can be discretised as:

$$L_{\text{loss}} = \sum_i m_i\, r_i\, v_i \sin\theta_i\ \left(1 - f_{\text{cap},i}\right). \tag{40}$$

A stochastic treatment can implement this discrete formulation by Monte Carlo sampling of $\rho(v,\theta)$: sample N particles from $\rho(v,\theta)$, assign each particle a mass $m_i = M_{\text{ej}}/N$, compute $h_i = r_i v_i \sin\theta_i$, compute $f_{\text{cap},i}$ from Eq. 39, and sum $L_{\text{loss}} = \sum_i m_i h_i (1 - f_{\text{cap},i})$. This Monte Carlo approach is recommended for 3D SPH validation (Limitation L2): it provides a robust check on the analytical estimates.

In a multi-phase formation scenario, ejecta are further grouped into k distinct formation phases:

$$L_{\text{loss}} = \sum_k \sum_i m_{k,i}\, r_{k,i}\, v_{k,i} \sin\theta_{k,i}\ \left(1 - f_{\text{cap},k,i}\right), \tag{41}$$

each phase k having its own mean ejection velocity $\bar{v}_k$ (decreasing as the proto-Earth loses mass and angular momentum), its own angular spread $\theta_{\max,k}$ (which may increase as the CMT reorganises between episodes), and its own mean capture efficiency $f_{\text{cap},k}$ (which increases as the proto-Moon grows, the positive feedback of Section 10.3).

11.5 Closure of the angular-momentum budget

The total angular-momentum budget is expressed as $L_{\text{init}} = L_{\text{remnant}} + L_{\text{accreted}} + L_{\text{esc}}$, with $L_{\text{accreted}} = \sum_i f_{\text{cap},i}\, h_i\, m_i$ and $L_{\text{esc}} = \sum_i (1 - f_{\text{cap},i})\, h_i\, m_i$. Using the parameters of the nominal TPT scenario (Table 5), we verify that the budget closes.

Table 3: Nominal parameters for closure of the angular-momentum budget.

Parameter	Symbol	Value
Proto-Earth initial moment of inertia	I_p	1.09×10^{38} kg m^2
Initial angular velocity	Ω_p	4.99×10^{-4} s^{-1}
Initial angular momentum	L_{init}	5.44×10^{34} J s
Mean ejection velocity	$\bar{v}$	9.8 km/s
Emission radius	r	7.9×10^{6} m
Number of episodes	N	3
Mass ejected per episode	M_{ej}	2.5×10^{22} kg
Capture efficiency (mean)	$\bar{f}_{\text{cap}}$	0.70
Final Earth angular momentum	L_{remnant}	5.86×10^{33} J s
Accreted lunar angular momentum	L_{accreted}	2.89×10^{34} J s
Escaped angular momentum	L_{esc}	1.96×10^{34} J s

The budget closes, at these nominal parameter values, with $L_{\text{remnant}} + L_{\text{accreted}} + L_{\text{esc}} = 5.44 \times 10^{34}\,\text{J s} = L_{\text{init}}$, a residual of $\sim 0.7\%$. This residual should be read as a check of *internal consistency* at nominal parameter values, not as an independent precision validation. Capture efficiency f_{cap} carries an estimated uncertainty of ± 0.05 (Table 6), which propagates to a $\sim 33\%$ uncertainty on L_{loss} via the sensitivity $1/(1 - f_{\text{cap}})$ derived in Section 15.3. A 0.7% residual at nominal values thus demonstrates that the model is self-consistent — the distributed ejecta model contains no hidden accounting error — but does not by itself demonstrate that the nominal parameters are correct to this precision. The relevant test is whether the budget continues to close, within the $\sim 5\%$ tolerance motivated by input-data uncertainties, when parameters are varied across their full range (Prediction P21).

11.6 Scaling laws and dimensionless form

The angular-momentum transport problem admits a natural non-dimensionalisation that reveals the essential physical dependences and simplifies the parameter space. We define the characteristic velocity scale $U_0 = U_{\text{crit}} \approx 9.8\,\text{km s}^{-1}$, the length scale $R_0 = R_p \approx 7.4 \times 10^{6}$ m, the mass scale $M_0 = M_{\text{ej}} \approx 2.5 \times 10^{22}$ kg, and the angular-momentum scale $H_0 = M_0 R_0 U_0 \approx 1.8 \times 10^{36}\,\text{kg m}^2\,\text{s}^{-1}$. The dimensionless variables $\tilde{v} = v/U_0$, $\tilde{r} = r/R_0$, $\tilde{m} = m/M_0$, and $\tilde{h} = h/(R_0 U_0) = \tilde{r}\tilde{v}\sin\theta$ recast the ejecta distribution as:

$$\tilde{\rho}(\tilde{v},\theta) = \tilde{M}_{\rm ej}\,\tilde{\mathcal{N}}_v\,\tilde{\mathcal{N}}_\theta\,\exp\left[-\frac{(\tilde{v}-\bar{\tilde{v}})^2}{2\tilde{\sigma}_v^2}\right]\cos\theta, \quad \tilde{v} > \tilde{v}_{\rm esc},\ |\theta| < \theta_{\rm max}, \tag{42}$$

with $\tilde{M}_{\rm ej} = 1$ by definition, $\bar{\tilde{v}} = \bar{v}/U_0 \approx 1$, $\tilde{\sigma}_v = \sigma_v/U_0 \approx 0.05$–$0.10$, and $\tilde{v}_{\rm esc} = v_{\rm esc}/U_0 \approx 1/\sqrt{2}$. The dimensionless capture efficiency takes the same erf form as Eq. 39, with $\tilde{v}_{\rm crit} \approx 1/\sqrt{2}$, and the dimensionless angular momentum lost, $L_{\rm loss}^{\tilde{}}$, recovers the physical quantity via $L_{\rm loss} = H_0\,L_{\rm loss}^{\tilde{}}$.

This dimensionless form reveals that the problem depends on only four dimensionless parameters: the mean ejection velocity relative to $U_{\rm crit}$ ($\bar{\tilde{v}} \approx 1$), the velocity spread ($\tilde{\sigma}_v \approx 0.05$–$0.10$), the torus's angular extent ($\theta_{\rm max} \approx 30°$), and the capture threshold ($\tilde{v}_{\rm crit} \approx 1/\sqrt{2}$). This low-dimensional parameter space keeps the model testable: each parameter can in principle be independently constrained by future simulations or observations.

The dimensionless form admits three asymptotic regimes. In the monokinetic limit ($\tilde{\sigma}_v \to 0$), the distribution reduces to a single velocity and recovers the single-velocity ballistic approximation used in earlier sections. In the broad-spectrum limit ($\tilde{\sigma}_v \gg 1$), the velocity distribution becomes uniform over the available range, maximising the angular momentum carried off by the fastest ejecta. In the sharp-cutoff limit ($\tilde{\sigma}_f \to 0$), capture efficiency becomes a step function, representing the idealised case where all material above the capture threshold is retained.

Falsifiable Prediction — P20 — Asymptotic-regime indicator

The actual TPT ejecta spectrum lies between the monokinetic and broad-spectrum limits, with $\tilde{\sigma}_v \approx 0.05$–$0.10$. The sharp-cutoff limit provides an upper bound on $L_{\rm loss}$, while the monokinetic limit provides a lower bound. The actual value, accessible via Monte Carlo simulation, should lie between these bounds.

Test: Monte Carlo sampling of $\rho(v,\theta)$ (Section 15.1) compared against the analytical monokinetic and broad-spectrum limits derived above.

The dimensionless form also allows scaling laws for $L_{\rm loss}$ to be derived in terms of the proto-Earth's fundamental parameters: $L_{\rm loss} \propto M_p R_p^2 \Omega_p$ (the standard scaling law for the angular momentum of a rotating body), $L_{\rm loss} \propto \sqrt{\sin\varepsilon}$ (from the obliquity dependence of the ejection-velocity threshold), and $\delta L_{\rm loss}/L_{\rm loss} \approx -\delta f_{\rm cap}/(1-f_{\rm cap})$. For $f_{\rm cap} = 0.70$, $1 - f_{\rm cap} = 0.30$, so a 10% variation in $f_{\rm cap}$ gives a 33% variation in $L_{\rm loss}$ — a sensitivity that underscores the importance of precisely determining $f_{\rm cap}$ in future simulations. Table 6 summarises the sensitivity of $L_{\rm loss}$ to each input parameter.

Table 4: Sensitivity of L_{loss} to input parameters.

Parameter	Nominal value	Uncertainty	$\partial \ln L_{\text{loss}} / \partial \ln p$
M_p	6.04×10^{24} kg	$\pm 5\%$	≈ 1
R_p	7.4×10^{6} m	$\pm 3\%$	≈ 2
Ω_p	4.99×10^{-4} s^{-1}	$\pm 10\%$	≈ 1
ε	55°	$\pm 10^\circ$	≈ 0.5
f_{cap}	0.70	± 0.05	≈ -3

The budget is most sensitive to f_{cap} (because of the $1/(1 - f_{\text{cap}})$ factor) and to R_p (because of the R_p^2 scaling). This sensitivity motivates the priority given to Limitation L2 (numerical determination of f_{cap}).

Falsifiable Prediction — P21 — Numerical closure of the angular-momentum budget

A Monte Carlo simulation of the ejecta spectrum (Section 11), with parameters sampled within their uncertainty ranges, should yield an angular-momentum budget that closes (Section 15.2) to within $\sim 5\%$.

Falsification: failure to close the angular-momentum budget to within $\sim 5\%$ would indicate either an error in the distribution function $\rho(v, \theta)$, or the need for an additional angular-momentum sink (e.g. tidal dissipation) not currently accounted for in the model.

The Monte Carlo implementation of this model is simple and computationally inexpensive; it is recommended as the first step of the 3D SPH validation programme, ahead of full hydrodynamic simulations.

12 Chronology: Moon formation in 3 to 4 weeks

12.1 The three phases of an episode

Each ejection episode comprises three phases: resonant acceleration, during which the CMT grows exponentially via the elliptical instability (Section 7.6), $t_{\text{acc}} = \ln(U_{\text{crit}}/U_{\text{turb}})/\sigma_{\text{res}} \approx 1.4$ day; ejection, during which the flow crosses the Roche radius ($3R_{\text{eq,proto}} \approx 22.2$ Mm) at U_{crit}, $t_{\text{ej}} \approx 0.6$ hour; and recharge, during which the CMT reorganises over 5–6 nutation periods, $t_{\text{rec}} \approx 10$–12 days. The Fe-Ni segregation rate (Stokes' law) is near-instantaneous between episodes ($\tau_{\text{Stokes}} \approx 254$ s, Appendix F), which implies that each accreted layer is chemically distinct from the previous one — a direct consequence underlying the seismic prediction P1.

12.2 Complete chronology

Phase	Duration	Cumulative
Episode 1: Acceleration	1.4 day	1.4 day
Episode 1: Ejection	0.6 h	1.4 day
Recharge (5–6 nutation periods)	10–12 days	11–13 days
Episode 2: Acceleration	1.4 day	12–14 days
Episode 2: Ejection	0.6 h	12–14 days
Recharge	10–12 days	22–26 days
Episode 3: Acceleration	1.4 day	23–27 days
Episode 3: Ejection	0.6 h	23–27 days
Total (N = 3)	**24–28 days**	

This 3–4 week timescale is the only intermediate temporal window between the giant impact (a few hours) and a synestia (~ 100 yr) on the Hf-W chronometer (half-life: 8.9 Myr). Episode 1 (acceleration plus ejection) forms the proto-POB, $R_1 \approx 1,560$ km. The recharge and second episode, $\approx$ 11–13 days, are asymmetric: transient post-ejection precession alters ε, producing the crustal dichotomy of Section 9.2. The recharge and third episode, again $\approx$ 11–13 days, reconstitute the lunar mass. The TPT window closes at $t \approx$ 30–50 Myr, and the Hadean dynamo switches on at $t \approx$ 330–350 Myr.

12.3 Age of the Moon: LMO solidification versus formation

The measured age (4.425 ± 0.025 Ga) is that of the solidification of the lunar magma ocean (LMO), not of formation. If the TPT is triggered at ≈ 4.467 Ga, LMO cooling (~ 40 Myr) gives a predicted solidification age,

$$t_{\mathrm{LMO}} \approx 4.467 - 0.040 = 4.427\,\mathrm{Ga}, \tag{43}$$

consistent to 0.1σ with the measured value. Prediction P8 is a formation age $>$ 4.45 Ga, directly testable by Artemis III.

13 Internal stratification and seismology programme

13.1 Architecture of concentric layers

Each ejected sheet originates from magma at a progressively more advanced stage of Fe-Ni segregation,

$$\left(\frac{\mathrm{Fe}}{\mathrm{Si}}\right)_{\mathrm{layer}\,k} > \left(\frac{\mathrm{Fe}}{\mathrm{Si}}\right)_{\mathrm{layer}\,k+1}, \tag{44}$$

since the source magma becomes progressively depleted in siderophile elements between episodes. Density contrasts between layers $\Delta\rho_{k,k+1} \approx 100$–$200\ \mathrm{kg\,m^{-3}}$ (Garcia et al., 2019) produce potentially detectable acoustic impedance contrasts.

13.2 Preservation window: 200–530 km

Two destructive processes bound the preservation window. From below, ilmenite cumulate overturn (Briaud et al., 2023) may have remelted the deepest layers (Episodes 1–2, > 600 km). From above, basaltic mare volcanism, active until ≈ 2 Ga (Li et al., 2021; Che et al., 2021), remelted the shallowest layers (< 100 km). The remaining preservation window therefore lies approximately between 200 and 530 km depth.

13.3 Interface depth: N=2 versus N=3

For $N = 2$ at equal masses, the interface between the two layers lies at:

$$d = R_{\mathrm{Moon}} - R_1 \approx 1,737 - 1,560 \approx 177\,\mathrm{km} \quad (N = 2). \tag{45}$$

For $N = 3$, the main interface lies within the 200–315 km window. Chang'e 7 seismic data will therefore allow discrimination between $N = 2$ and $N = 3$.

14 Observational constraints and existing evidence

14.1 Isotopic identity Δ^{17}O < 5 ppm

The Earth–Moon isotopic composition is nearly identical for O, Cr, and Ti, established to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The TPT satisfies this constraint *mechanically*: at each ejection episode, material is drawn from the contemporary Hadean terrestrial mantle, with no exogenous isotopic source. No *ad hoc* isotopic mixing is required.

14.2 Iron depletion

The Moon is significantly less iron-rich than Earth's mantle (O'Neill, 1991; Jones & Drake, 1986). In the TPT, this depletion is a direct consequence of the single engine: at each successive episode, the ejected mantle is more depleted in siderophiles than the previous one, Fe-Ni segregation progressively removing iron from the source zone between episodes.

14.3 Angular momentum

The Earth–Moon system's angular-momentum budget requires a rapidly rotating proto-Earth. The value $T_{\mathrm{rot}} = 3.5$ h is consistent with the median value from N-body accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010): rapid rotation is the norm at the end of accretion, not an exception.

14.4 Chang'e-6 results: preliminary observational support

The Chang'e-6 mission returned 1,935 g of samples from the Apollo basin, within the South Pole–Aitken (SPA) region, in June 2024. Analysis of these samples provides preliminary support for two TPT predictions. Yue et al. (2026) dated Chang'e-6 norite samples to 4247 ± 5 Myr by the Pb-Pb method. These norites are interpreted as differentiated products of the SPA impact melt sheet, representing deep lunar mantle material excavated by the SPA impact. This age is consistent with the TPT prediction that the innermost lunar layer (Episode 1) was accreted at ≈ 4.5 Ga and partially reset by the SPA impact at ≈ 4.25 Ga. Xu et al. (2025) further report olivine fragments in Chang'e-6 samples containing Fe and Mn concentrations significantly higher than those found in standard lunar mantle, with zinc concentrations $\approx 100\times$ higher than expected. Although some of this enrichment is attributed to meteoritic contamination, the olivine fragments themselves formed through mixing of lunar rock and molten asteroidal material. The high iron content of material excavated from SPA depths is *consistent* with the Fe/Si gradient increasing with depth (P3) and provides preliminary support for the South Pole Fe/Si anomaly (P6), detailed below.

15 Falsifiable predictions

orbital mechanics. The nominal value $f_{\text{cap}} \approx 0.70$, independently estimated from the CMT's cohesive fraction and re-accretion of co-orbital fragments, is consistent with the mass-weighted average of this function over the proposed ejecta spectrum.

15.1 Discrete formulation and Monte Carlo implementation

For numerical implementation, the expression can be discretised as:

$$L_{\text{loss}} = \sum_i m_i\, r_i\, v_i \sin\theta_i\, \left(1 - f_{\text{cap},i}\right). \tag{46}$$

A stochastic treatment can implement this discrete formulation via Monte Carlo sampling of $\rho(v, \theta)$: sample N particles from $\rho(v, \theta)$, assign each particle a mass $m_i = M_{\text{ej}}/N$, compute $h_i = r_i v_i \sin\theta_i$, compute $f_{\text{cap},i}$ from Eq. 39, and sum $L_{\text{loss}} = \sum_i m_i h_i (1 - f_{\text{cap},i})$. This Monte Carlo approach is recommended for 3D SPH validation (Limitation L2): it provides a robust check on the analytical estimates.

In a multi-phase formation scenario, ejecta are further grouped into k distinct formation phases:

$$L_{\text{loss}} = \sum_k \sum_i m_{k,i}\, r_{k,i}\, v_{k,i} \sin\theta_{k,i}\, \left(1 - f_{\text{cap},k,i}\right), \tag{47}$$

each phase k having its own mean ejection velocity $\bar{v}_k$ (decreasing as the proto-Earth loses mass and angular momentum), its own angular spread $\theta_{\max,k}$ (which may increase as the CMT reorganises between episodes), and its own mean capture efficiency $f_{\text{cap},k}$ (which increases as the proto-Moon grows, the positive feedback of Section 10.3).

15.2 Closure of the angular-momentum budget

The total angular-momentum budget is expressed as $L_{\text{init}} = L_{\text{remnant}} + L_{\text{accreted}} + L_{\text{esc}}$, with $L_{\text{accreted}} = \sum_i f_{\text{cap},i}\, h_i\, m_i$ and $L_{\text{esc}} = \sum_i (1 - f_{\text{cap},i})\, h_i\, m_i$. Using the nominal TPT scenario parameters (Table 5), we verify that the budget closes.

Table 5: Nominal parameters for angular-momentum budget closure.

Parameter	Symbol	Value
Proto-Earth initial moment of inertia	I_p	1.09×10^{38} kg m^2
Initial angular velocity	Ω_p	4.99×10^{-4} s^{-1}
Initial angular momentum	L_{init}	5.44×10^{34} J s
Mean ejection velocity	$\bar{v}$	9.8 km/s
Emission radius	r	7.9×10^{6} m
Number of episodes	N	3
Mass ejected per episode	M_{ej}	2.5×10^{22} kg
Capture efficiency (mean)	$\bar{f}_{\text{cap}}$	0.70
Earth's final angular momentum	L_{remnant}	5.86×10^{33} J s
Accreted lunar angular momentum	L_{accreted}	2.89×10^{34} J s
Escaped angular momentum	L_{esc}	1.96×10^{34} J s

The budget closes, at these nominal parameter values, with $L_{\text{remnant}} + L_{\text{accreted}} + L_{\text{esc}} = 5.44 \times 10^{34}\,\text{J s} = L_{\text{init}}$, a residual of $\sim 0.7\%$. This residual should be read as an *internal consistency* check at nominal parameter values, not as an independent validation of precision. Capture efficiency f_{cap} carries an estimated uncertainty of ± 0.05 (Table 6), which propagates into a $\sim$ 33% uncertainty on L_{loss} via the $1/(1 - f_{\text{cap}})$ sensitivity derived in Section 15.3. A 0.7% residual at nominal values thus demonstrates that the model is self-consistent — the distributed ejecta model contains no hidden bookkeeping error — but does not by itself demonstrate that the nominal parameters are correct to this precision. The relevant test is whether the budget continues to close, within the $\sim$ 5% tolerance motivated by input-data uncertainties, when parameters are varied across their full range (Prediction P21).

15.3 Scaling laws and non-dimensional form

The angular-momentum transport problem admits a natural non-dimensionalisation that reveals the essential physical dependencies and simplifies the parameter space. We define the characteristic velocity scale $U_0 = U_{\rm crit} \approx 9.8\ \rm km\,s^{-1}$, the length scale $R_0 = R_p \approx 7.4 \times 10^6$ m, the mass scale $M_0 = M_{\rm ej} \approx 2.5 \times 10^{22}$ kg, and the angular-momentum scale $H_0 = M_0 R_0 U_0 \approx 1.8 \times 10^{36}\ \rm kg\,m^2\,s^{-1}$. The non-dimensional variables $\tilde{v} = v/U_0$, $\tilde{r} = r/R_0$, $\tilde{m} = m/M_0$, and $\tilde{h} = h/(R_0 U_0) = \tilde{r}\tilde{v}\sin\theta$ recast the ejecta distribution as:

$$\tilde{\rho}(\tilde{v},\theta) = \tilde{M}_{\rm ej}\,\tilde{\mathcal{N}}_v\,\tilde{\mathcal{N}}_\theta\,\exp\left[-\frac{(\tilde{v}-\bar{\tilde{v}})^2}{2\tilde{\sigma}_v^2}\right]\cos\theta, \quad \tilde{v} > \tilde{v}_{\rm esc},\ |\theta| < \theta_{\rm max}, \tag{48}$$

with $\tilde{M}_{\rm ej} = 1$ by definition, $\bar{\tilde{v}} = \bar{v}/U_0 \approx 1$, $\tilde{\sigma}_v = \sigma_v/U_0 \approx 0.05$–$0.10$, and $\tilde{v}_{\rm esc} = v_{\rm esc}/U_0 \approx 1/\sqrt{2}$. The non-dimensional capture efficiency takes the same erf form as Eq. 39, with $\tilde{v}_{\rm crit} \approx 1/\sqrt{2}$, and the non-dimensional angular momentum lost, $\tilde{L_{\rm loss}}$, recovers the physical quantity via $L_{\rm loss} = H_0\,\tilde{L_{\rm loss}}$.

This non-dimensional form reveals that the problem depends only on four non-dimensional parameters: the mean ejection velocity relative to $U_{\rm crit}$ ($\bar{\tilde{v}} \approx 1$), the velocity spread ($\tilde{\sigma}_v \approx 0.05$–$0.10$), the angular extent of the torus ($\theta_{\rm max} \approx 30^\circ$), and the capture threshold ($\tilde{v}_{\rm crit} \approx 1/\sqrt{2}$). This low-dimensional parameter space keeps the model testable: each parameter can in principle be independently constrained by future simulations or observations.

The non-dimensional form admits three asymptotic regimes. In the monokinetic limit ($\tilde{\sigma}_v \to 0$), the distribution reduces to a single velocity and recovers the single-velocity ballistic approximation used in earlier sections. In the broad-spectrum limit ($\tilde{\sigma}_v \gg 1$), the velocity distribution becomes uniform over the available range, maximising the angular momentum carried by the fastest ejecta. In the sharp-cutoff limit ($\tilde{\sigma}_f \to 0$), capture efficiency becomes a step function, representing the idealised case where all material above the capture threshold is retained.

Falsifiable Prediction — P20 — Asymptotic regime indicator

The TPT's actual ejecta spectrum lies between the monokinetic and broad-spectrum limits, with $\tilde{\sigma}_v \approx 0.05$–$0.10$. The sharp-cutoff limit provides an upper bound on $L_{\rm loss}$, while the monokinetic limit provides a lower bound. The actual value, accessible through Monte Carlo simulation, should lie between these bounds.

Test: Monte Carlo sampling of $\rho(v,\theta)$ (Section 15.1) compared with the analytical monokinetic and broad-spectrum limits derived above.

The non-dimensional form also allows scaling laws for $L_{\rm loss}$ to be derived as a function of the proto-Earth's fundamental parameters: $L_{\rm loss} \propto M_p R_p^2 \Omega_p$ (the standard scaling law for the angular momentum of a rotating body), $L_{\rm loss} \propto \sqrt{\sin\varepsilon}$ (from the obliquity dependence of the ejection velocity threshold), and $\delta L_{\rm loss}/L_{\rm loss} \approx -\delta f_{\rm cap}/(1-f_{\rm cap})$. For $f_{\rm cap} = 0.70$, $1 - f_{\rm cap} = 0.30$, so a 10% variation in $f_{\rm cap}$ gives a 33% variation in $L_{\rm loss}$ — a sensitivity that underlines the importance of precisely determining $f_{\rm cap}$ in future simulations. Table 6 summarises the sensitivity of $L_{\rm loss}$ to each input parameter.

Table 6: Sensitivity of $L_{\rm loss}$ to input parameters.

Parameter	Nominal value	Uncertainty	$\partial \ln L_{\rm loss}/\partial \ln p$
M_p	6.04×10^{24} kg	$\pm 5\%$	≈ 1
R_p	7.4×10^{6} m	$\pm 3\%$	≈ 2
Ω_p	4.99×10^{-4} s^{-1}	$\pm 10\%$	≈ 1
ε	55°	$\pm 10^\circ$	≈ 0.5
$f_{\rm cap}$	0.70	± 0.05	≈ -3

The budget is most sensitive to $f_{\rm cap}$ (due to the $1/(1-f_{\rm cap})$ factor) and to R_p (due to the R_p^2 scaling law). This sensitivity motivates the priority given to Limitation L2 (numerical determination of $f_{\rm cap}$).

Falsifiable Prediction — P21 — Numerical closure of the angular-momentum budget

A Monte Carlo simulation of the ejecta spectrum (Section 11), with parameters sampled across their uncertainty ranges, should give an angular-momentum budget that closes (Section 15.2) to within $\sim 5\%$.

Falsification: failure to close the angular-momentum budget to within $\sim 5\%$ would indicate either an error in the distribution function $\rho(v, \theta)$, or the need for an additional angular-momentum sink (e.g. tidal dissipation) not currently accounted for in the model.

The Monte Carlo implementation of this model is simple and computationally inexpensive; it is recommended as the first step of the 3D SPH validation programme, ahead of full hydrodynamic simulations.

16 Chronology: lunar formation in 3 to 4 weeks

16.1 The three phases of an episode

Each ejection episode comprises three phases: resonant acceleration, during which the CMT grows exponentially via the elliptical instability (Section 7.6), $t_{\rm acc} = \ln(U_{\rm crit}/U_{\rm turb})/\sigma_{\rm res} \approx 1.4$ days; ejection, during which the flow crosses the Roche radius ($3R_{\text{eq,proto}} \approx 22.2$ Mm) at U_{crit}, $t_{\text{ej}} \approx 0.6$ hours; and recharge, during which the CMT reorganises over 5–6 nutation periods, $t_{\text{rec}} \approx 10$–12 days. The Fe-Ni segregation rate (Stokes' law) is near-instantaneous between episodes ($\tau_{\text{Stokes}} \approx 254$ s, Appendix F), implying that each accreted layer is chemically distinct from the previous one — a direct consequence underpinning the seismic prediction P1.

16.2 Complete chronology

Phase	Duration	Cumulative
Episode 1: Acceleration	1.4 days	1.4 days
Episode 1: Ejection	0.6 h	1.4 days
Recharge (5–6 nutation periods)	10–12 days	11–13 days
Episode 2: Acceleration	1.4 days	12–14 days
Episode 2: Ejection	0.6 h	12–14 days
Recharge	10–12 days	22–26 days
Episode 3: Acceleration	1.4 days	23–27 days
Episode 3: Ejection	0.6 h	23–27 days
Total (N = 3)	**24–28 days**	

This 3–4 week timescale is the only intermediate temporal window between the giant impact (a few hours) and a synestia ($\sim$ 100 yr) on the Hf-W chronometer (half-life: 8.9 Myr). Episode 1 (acceleration plus ejection) forms the proto-POB, $R_1 \approx 1,560$ km. Recharge and the second episode, $\approx$ 11–13 days, are asymmetric: the transient post-ejection precession shifts ε, producing the crustal dichotomy of Section 9.2. Recharge and the third episode, again $\approx$ 11–13 days, restore the lunar mass. The TPT window closes at $t \approx 30$–50 Myr, and the Hadean dynamo switches on at $t \approx 330$–350 Myr.

16.3 Age of the Moon: LMO solidification versus formation

The measured age (4.425 ± 0.025 Ga) is that of lunar magma ocean (LMO) solidification, not formation. If the TPT is triggered at ≈ 4.467 Ga, LMO cooling ($\sim$ 40 Myr) gives a predicted solidification age,

$$t_{\text{LMO}} \approx 4.467 - 0.040 = 4.427\,\text{Ga}, \tag{49}$$

consistent to 0.1σ with the measured value. Prediction P8 is a formation age $>$ 4.45 Ga, directly testable by Artemis III.

17 Internal stratification and seismology programme

17.1 Architecture of concentric layers

Each ejected sheet originates from magma at a progressively more advanced stage of Fe-Ni segregation,

$$\left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer}\,k} > \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer}\,k+1}, \tag{50}$$

since the source magma becomes progressively depleted in siderophile elements between episodes. Density contrasts between layers $\Delta\rho_{k,k+1} \approx 100$–$200\ \text{kg}\,\text{m}^{-3}$ (Garcia et al., 2019) produce potentially detectable acoustic impedance contrasts.

17.2 Preservation window: 200–530 km

Two destructive processes bound the preservation window. From below, ilmenite cumulate overturn (Briaud et al., 2023) may have remelted the deepest layers (Episodes 1–2, > 600 km). From above, mare basaltic volcanism, active until ≈ 2 Ga (Li et al., 2021; Che et al., 2021), remelted the shallowest layers (< 100 km). The remaining preservation window thus lies approximately between 200 and 530 km depth.

17.3 Interface depth: N=2 versus N=3

For $N = 2$ at equal masses, the interface between the two layers lies at:

$$d = R_{\text{Moon}} - R_1 \approx 1{,}737 - 1{,}560 \approx 177\,\text{km} \quad (N = 2). \tag{51}$$

For $N = 3$, the main interface lies within the 200–315 km window. Chang'e 7 seismic data will therefore allow discrimination between $N = 2$ and $N = 3$.

18 Observational constraints and existing evidence

18.1 Isotopic identity Δ^{17}O < 5 ppm

The Earth–Moon isotopic composition is nearly identical for O, Cr, and Ti, established to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The TPT satisfies this constraint *mechanically*: at each ejection episode, material is drawn from the contemporaneous Hadean terrestrial mantle, with no exogenous isotopic source. No *ad hoc* isotopic mixing is required.

18.2 Iron depletion

The Moon is significantly less iron-rich than Earth's mantle (O'Neill, 1991; Jones & Drake, 1986). In the TPT, this depletion is a direct consequence of the single engine: at each successive episode, the ejected mantle is more depleted in siderophiles than the previous one, Fe-Ni segregation progressively removing iron from the source zone between episodes.

18.3 Angular momentum

The Earth–Moon system's angular-momentum budget requires a rapidly rotating proto-Earth. The value $T_{\text{rot}} = 3.5$ h is consistent with the median value of N-body accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010): rapid rotation is the norm at the end of accretion, not an exception.

18.4 Chang'e-6 results: preliminary observational support

The Chang'e-6 mission returned 1,935 g of samples from the Apollo basin, within the South Pole–Aitken (SPA) region, in June 2024. Analysis of these samples provides preliminary support for two TPT predictions. Yue et al. (2026) dated Chang'e-6 norite samples at 4247 ± 5 Myr by the Pb-Pb method. These norites are interpreted as differentiated products of the SPA impact melt sheet, representing deep lunar-mantle material excavated by the SPA impact. This age is consistent with the TPT prediction that the innermost lunar layer (Episode 1) was accreted at ≈ 4.5 Ga and partially reset by the SPA impact at ≈ 4.25 Ga. Xu et al. (2025) further report olivine fragments in Chang'e-6 samples containing Fe and Mn concentrations significantly higher than those found in standard lunar mantle, with zinc concentrations $\approx 100\times$ higher than expected. Although some of this enrichment is attributed to meteoritic contamination, the olivine fragments themselves formed through mixing of lunar rock and molten asteroidal material. The high iron content of material excavated from the SPA depths is *consistent* with the Fe/Si gradient increasing with depth (P3) and provides preliminary support for the South Pole Fe/Si anomaly (P6), detailed below.

19 Falsifiable predictions

This section catalogues every quantitative, falsifiable prediction made by the TPT. Each entry states the prediction, its physical derivation where this has not already been given in full elsewhere in the manuscript, the relevant test, and a falsification threshold set explicitly in advance of observational data.

Falsifiable Prediction — P1 — Seismic interface at $d \approx 200$–315 km (Priority 1)

Concentric stratification produces acoustic impedance contrasts $|R| \in [0.01, 0.04]$ between layers. For $N = 2$, the interface lies at $d \approx 177$ km. For $N = 3$, the main interface lies at $d \approx 200$–315 km.

Detectability with Chang'e-7 depends on seismometer sensitivity (target $10^{-8}\,\text{m}\,\text{s}^{-2}\,\text{Hz}^{-1/2}$ at 0.1–1 Hz). This sensitivity should resolve $|R| > 0.01$, provided at least 10 events of $M > 2.5$ are recorded during the first 6 months of operation.

Falsification: no phase conversion detected within the 200–530 km window after deployment of at least 3 seismic stations and detection of at least 10 lunar events of

$M > 2.5$.

Missions: Chang'e 7 (August 2026, South Pole seismometer), FSS (Farside Seismic Suite), LEMS (Lunar Environment Monitoring Station), Artemis III (2028–2029).

Falsifiable Prediction — P2 — Nearside versus farside seismic asymmetry

The asymmetric second episode produces a sharper, deeper interface beneath the farside.

Falsification: spherical symmetry confirmed by a complete tomographic inversion.

Falsifiable Prediction — P3 — Fe/Si gradient increasing with depth

The Fe/Si ratio increases with depth in the lunar mantle, reflecting the progressive Fe-Ni depletion of the source magma between episodes. Quantitatively, with the total cumulative fraction of Fe removed from the mantle $\xi_{\rm Fe}^{\rm sat} \approx 0.15$ (Rubie et al., 2015) and $N = 3$ segregation episodes, the fraction of Fe removed per episode, assuming a geometric progression over the N episodes, is:

$$\xi_{\rm Fe,episode} = 1 - (1 - \xi_{\rm Fe}^{\rm sat})^{1/N} \approx 1 - (0.85)^{1/3} \approx 0.053, \tag{52}$$

i.e. $\approx 5.3\%$ of the remaining Fe removed at each episode. The resulting Fe/Si ratio between adjacent layers is multiplicative:

$$\frac{({\rm Fe/Si})_{{\rm Ep}k}}{({\rm Fe/Si})_{{\rm Ep}(k+1)}} = \frac{1}{1 - \xi_{\rm Fe,episode}} \approx 1.056, \tag{53}$$

giving an enrichment of about 5.6% between Episode 2 and Episode 3, and a cumulative enrichment $({\rm Fe/Si})_{\rm Ep1}/({\rm Fe/Si})_{\rm Ep3} = (1 - \xi_{\rm Fe,episode})^{-2} \approx 1.115$, i.e. a total enrichment of about 11.5% between Episode 1 (the deepest layer, richest in Fe) and Episode 3 (the shallowest). The SPA basin preferentially exposes Episode 1 material.

Preliminary support: the high-Fe/Mn olivines from Chang'e-6 (Xu et al., 2025). **Missions:** Chang'e-6 (further analyses), Artemis III.

Falsifiable Prediction — P4 — Dynamo delay: 290–360 Myr

Progressive dynamo emergence predicts continuous paleointensity growth from 4.55 to 4.20 Ga, with no abrupt onset (Tarduno et al., 2025).

Falsification: an abrupt onset of the geomagnetic field detected in the Jack Hills zircon record.

Falsifiable Prediction — P5 — Instability window: $\varepsilon \in [57°, 70°]$

The parametric elliptical instability requires ε within this range.

Test: N-body simulations of terrestrial protoplanet obliquity.

Falsifiable Prediction — P6 — South Pole Fe/Si anomaly (derived from P3 and the South Pole–Aitken impact simulations of Wakita et al. 2026)

Observational context. The hydrodynamic simulations of Wakita et al. (2026) establish three facts independent of the TPT: the SPA impact excavated lunar mantle material from depths greater than 90 km; the impactor's oblique, north-south trajectory produced an asymmetric ejecta plume that preferentially concentrated deep material toward the South Pole; and about 350 m of mantle material is predicted at the surface near Shackleton crater on average.

TPT interpretation. Within the TPT framework, the lunar mantle is stratified into concentric layers (P1), with Fe/Si decreasing from the deepest layer (Episode 1) to the shallowest (Episode 3), as quantitatively derived in P3 above. The combined thickness of the Episode 2 and Episode 3 layers is about 315 km (Section 17). Material excavated by the SPA impact from depths greater than 90 km, and concentrated toward the South Pole, therefore originates preferentially from Episode 2 and, at greater excavation depths, from Episode 1.

Quantitative scenario. Wakita et al. (2026) do not provide a quantified fraction of Episode 1 versus Episode 2 material in the SPA ejecta plume, since their simulation does not incorporate TPT stratigraphy; this fraction must therefore be bounded from first principles rather than read directly from their results. We consider the worst-case scenario compatible with the TPT: SPA excavation reaches only ≈ 100 km (barely beyond the 90 km minimum established by Wakita et al. 2026), so that only Episode 2 material — not the deeper, more strongly enriched Episode 1 — contributes to the ejecta, and we conservatively take the minimum plausible fraction of Episode 2 material in the plume to be 30%. With the $\approx 5.6\%$ Fe/Si enrichment of Episode 2 relative to Episode 3 derived in Eq. 53, this minimal scenario predicts an enrichment of:

$$\Delta(\mathrm{Fe/Si})_{\min} \approx 0.30 \times 0.056 \approx 0.017 \approx 1.7\%. \tag{54}$$

If excavation instead reaches beyond ≈ 315 km, Episode 1 material contributes directly; by way of illustration, a mixture of 30% Episode 1 material, 30% Episode 2, and 40% Episode 3 would give $\Delta(\mathrm{Fe/Si}) \approx 0.30 \times 0.115 + 0.30 \times 0.056 \approx 5.1\%$, a value that requires a non-negligible Episode 1 contribution to be reached.

Quantitative prediction. South Pole samples (Chang'e-7, Artemis III) should show a higher Fe/Si ratio than the average of equatorial samples (Apollo) and nearside samples (Chang'e-5), with an enrichment of at least $\approx 1.7\%$ in the minimal scenario above and potentially up to $\approx 11.5\%$ if Episode 1 material dominates the sampled ejecta.

Falsification threshold. We set, in advance of any observational data, the falsification threshold at:

$$(\mathrm{Fe/Si})_{\mathrm{SPA}} \leq 1.015 \times (\mathrm{Fe/Si})_{\text{lunar average}} \implies \text{P6 falsified,} \tag{55}$$

i.e. an enrichment below 1.5%, set slightly under the 1.7% minimal-scenario value of Eq. 54 to allow for experimental uncertainty without weakening the substance of the test: even the most conservative scenario compatible with the TPT predicts an enrichment above this threshold, so that a measured enrichment below it would be incompatible with the theory as constructed here. An enrichment between 1.5% and 5.6% (the maximum attainable by Episode 2 material alone, from Eq. 53 applied to a pure Episode 2 sample) is read as partial support, consistent with Episode 2-dominated ejecta without requiring an Episode 1 contribution. An enrichment above 5.6% cannot be produced by Episode 2 and 3 material alone, and therefore constitutes direct evidence of an Episode 1 contribution, providing strong support for the deep stratigraphy predicted by P1 and P3.

Epistemic note on the grading bounds. The lower bound (1.5%) rests on an estimate of the minimum fraction of Episode 2 material in SPA ejecta (30%), which is a modelling assumption. The upper bound (5.6%) is a physical constraint: it is the maximum enrichment attainable by a pure Episode 2–Episode 3 mixture, with no Episode 1 contribution. Consequently, any measured enrichment above 5.6% constitutes definitive evidence of Episode 1 material, a qualitatively different regime from the lower-bound scenario. The two bounds are therefore not of the same epistemic nature; this asymmetry is explicitly acknowledged here.

Test: Chang'e-7 surface analysers and Artemis III returned samples from the South Pole, compared with existing Fe/Si measurements from Apollo (equatorial) and Chang'e-5 (nearside).

Acknowledged limitation: this prediction shares Limitation L9 (impact mixing in SPA ejecta) with prediction P3, and Limitation L11 (the 30% Episode 2 fraction itself) below.

Falsifiable Prediction — P7 — Signature in the Hf-W chronometer

A 3–4 week formation duration is the only intermediate temporal window between the giant impact (a few hours) and a synestia (~ 100 yr) on the Hf-W chronometer (half-life: 8.9 Myr). The ^{182}Hf/^{180}Hf ratio in Episode 1 samples should correspond to a differentiation age of 4.467 Ga (100 ± 10 Myr after the CAIs), versus 60 ± 10 Myr for the giant impact (Kleine et al., 2002).

Mission: Artemis III (2028–2029).

Falsifiable Prediction — P8 — Formation age > 4.45 Ga

The measured age of 4.425 Ga is that of LMO solidification, not formation (Section 16). The formation age is about 4.467 Ga.

Test: direct dating of Episode 1 rocks exposed on the central peaks of the SPA basin (Artemis III).

Falsifiable Prediction — P9 — Mare basalt compatibility with Layer 1

The melting depth of mare basalts (200–500 km, ± 100 km) is compatible with a source within layer 1 ($> d_{\mathrm{P1}}$), although this cannot be confirmed without more precise geochemical data.

Test: high-resolution isotopic geochemistry of Apollo and Chang'e-5 mare basalts (Li et al., 2021).

Falsifiable Prediction — P10 — Seismic interface at $d \approx 177$ km for $N = 2$

For $N = 2$ episodes of equal mass, the single interface between the two concentric layers lies at $d \approx 177$ km. This depth lies outside the main preservation window (200–530 km) as it has been partially remelted by mare volcanism and/or ilmenite cumulate overturn. However, partial remnants might be detected as weak reflectors or seismic-velocity variations in the 150–200 km zone.

Predictions P17 (capture efficiency, Section 10.3), P20 (asymptotic regime indicator, Section 15.3), P21 (numerical closure of the angular-momentum budget, Section 15.2), and P22 (the unified criterion for double-well emergence, Section 9.2.2) are derived in full, with their own falsification conditions, at the points in the manuscript indicated above, rather than repeated here.

20 Validation windows: three imminent tests

20.1 Chang'e 7: imminent seismic validation (August 2026)

The Chang'e 7 mission (CNSA, scheduled launch August 2026) will deploy the first seismometer at the lunar South Pole since Apollo 17 (1972). Its data will provide a direct test of P1, P2, and P6. The prediction is pre-registered and quantified: $d \approx 200$–315 km, $|R| \in [0.01, 0.04]$ for the seismic interface, and an enrichment above 1.5% in Fe/Si for South Pole ejecta. The falsification criteria are explicit in each case.

20.2 The South Pole–Aitken basin: a stratigraphic revealer

The SPA basin did not create the concentric lunar stratigraphy: it *excavated* it. The layered structure predates the SPA impact at ≈ 4.25 Ga (Yue et al., 2026), encoded during TPT differentiation at ≈ 4.5 Ga. The impact is the accidental revealer of a far older internal architecture.

High-resolution 3D simulations (Wakita et al., 2026) show that the oblique north-south trajectory of the SPA impactor preferentially launched deep-mantle ejecta toward the South Pole in a butterfly-shaped plume — precisely the Artemis III landing zone. These same simulations predict ≈ 350 m of mantle material on average at this site, excavated from depths greater than 90 km, consistent both with the TPT preservation window and with the quantitative scenario underlying prediction P6.

20.3 Artemis III: direct sampling of Episode 1 material

The Artemis III mission (NASA, scheduled for 2028–2029) will land astronauts near Shackleton crater, at the South Pole. According to Wakita et al. (2026), surface samples at this location could include deep-mantle ejecta from the SPA impact. If so, these samples will allow a direct test of P3, P6, P7, and P8: their Fe/Si ratio, their crystallisation age (≈ 4.467 Ga for primary Episode 1 material), and their isotopic signature ($\Delta^{17}O < 5$ ppm relative to the contemporaneous terrestrial mantle) are all quantitative TPT predictions.

Chang'e 7 (2026) targets seismology and the Fe/Si anomaly (P1, P2, P6), while Artemis III (2028) targets geochemistry and chronology (P3, P6, P7, P8). Two independent space agencies converge on a single geographic site — the lunar South Pole — to test a single internal architecture.

21 Discussion: main objections and responses

21.1 Comparison with the giant-impact model

Constraint	Giant impact	TPT (this work)
Isotopic identity	Requires very specific impactor composition and mixing conditions	Satisfied mechanically by construction
Iron depletion	Requires a pre-differentiated impactor (Theia)	Direct consequence of the single engine
Crustal dichotomy	Not naturally predicted	Qualitative prediction: post-Episode 2 precession (L7)
Dynamo delay	Not predicted	Proposed: $\approx$ 350 Myr, with no free parameter (L4)
Seismic predictions	None	P1 testable by Chang'e 7 (August 2026)
Chang'e-6 Fe anomaly	Not predicted	Consistent with P3 and P6 (Xu et al., 2025; Yue et al., 2026)
Formation timescale	A few hours	3–4 weeks
Hf-W signature	60 ± 10 Myr after CAIs	100 ± 10 Myr after CAIs (P7)

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in lunar formation (Sossi et al., 2025). This conclusion motivates the exploration of alternatives grounded in the internal dynamics of the proto-Earth.

21.2 Responses to the main objections

Objection 1: "A large number of unjustified assumptions." Hypotheses are explicitly ranked into three levels (Section 8, Table 2): Level 1 (established) denotes results from the literature or closed-form derivations (e.g. $E_{\text{grav}}/E_{\text{fus}} \approx 155$, the Bingham-Herschel law); Level 2 (constrained) denotes physically motivated hypotheses not yet numerically validated (e.g. $\varepsilon \in [40°, 70°]$, the Kramers rate); Level 3 (speculative) denotes hypotheses awaiting 3D SPH validation (e.g. the emergence of a single $\ell = 1, m = 1$ torus, the exact value of f_{cap}, the quantitative validity of the Gaussian approximation discussed below). The TPT does not rest on all these hypotheses with equal weight. Observational predictions (Section 19) are independent of Level-3 hypotheses.

Objection 2: "The CMT has not been demonstrated." In the rapid-rotation limit (Ro $\ll 1$), the azimuthal component of the Navier-Stokes equation for a Bingham-Herschel fluid reduces to a potential-vorticity conservation equation. Linearising about the mean flow near the instantaneous dynamical equator ($\phi = 0$, perpendicular to $\boldsymbol{\Omega}(t)$, as defined in Section 3), and applying the standard multiple-scales expansion for Rossby-wave envelopes (Redekopp, 1977; Luo, 1991; Yin et al., 2019), one obtains a nonlinear Schrödinger equation for the envelope $A(y, t)$, where $V_{\text{eff}}(y)$ is the sum of the potentials listed in Table 1. A stationary ansatz $A(y, t) = \psi(y)e^{-iEt/\hbar_{\text{eff}}}$ then gives a time-independent nonlinear Schrödinger equation for $\psi(y)$, the nonlinearity arising from the advective term of the potential-vorticity equation. The ground-state mode of the *linearised* version of this equation, with $V_{\text{eff}}(y)$ expanded to quadratic order around $y = 0$, is a Gaussian wavefunction centred on the instantaneous equator, confined to the band $|\phi| < 30°$ — a direct consequence of the fact that the seven potentials of Table 1 all have their extrema at $y = 0$. Whether this Gaussian remains a quantitatively adequate approximation of the ground state of the full nonlinear equation, rather than merely its qualitative first-order limit, has not been established for this system and is acknowledged as Limitation L12 (Section 22). The complete derivation is given in Appendix C. Quantitative validation of the single-torus structure further awaits 3D SPH simulation (Limitation L1).

Objection 3: "The double-well potential is tuned to give $N = 2$–3 episodes." Section 9.2.2 shows that the double well emerges from the coupled system of Eqs. 28–29. The emergence condition is a surface in the $(\Omega, |g_{\text{eff}}|)$ plane (Prediction P22). The number $N = 2$–3 is a consequence of barrier decay, not an a priori assumption.

Objection 4: "$f_{\text{cap}} = 0.70$ is high compared with accretion simulations (0.3–0.5)." The classical range (0.3–0.5) concerns impacts between solid bodies. Here, the ejected material is magma at 3,000–4,000 K, with viscous coalescence and toroidal orbital confinement. Appendix G establishes a physical lower bound $f_{\text{cap}}^{\min} \gtrsim 0.45$–0.55, independent of the BH cohesion hypothesis. The value 0.70 falls within a mechanically justifiable range.

Objection 5: "Are the thresholds required for double-well emergence physically attainable?" At the proto-Earth's nominal global parameters, the criterion $\Delta > 0$ requires a reduction of $|g_{\text{eff}}|$ of about 91% at fixed Ω (Appendix C, Section C.4). Several complementary physical mechanisms, specific to the CMT's local dynamics and to the Hadean environment, together contribute to closing this gap (Appendix H): the CMT's own effective density, reduced more strongly than the proto-Earth's overall density through segregation extending beyond the Fe-Ni couple alone and through the vesicularity of the *ʻaā*-type magmatic flow; obliquity evaluated at its geometric maximum, consistent with the absence of a preferred obliquity value prior to orbital stabilisation; and the angular-momentum contribution from continuous planetesimal bombardment. Taken together, these mechanisms render the criterion $\Delta > 0$ attainable within the active TPT window.

Objection 6: "The manuscript is too speculative for review." The TPT is submitted as a *theory with falsifiable predictions*, not as an observational result. Validation will proceed in two phases: Phase 1 (2026–2029), observational tests by Chang'e 7 (August 2026) and Artemis III (2028–2029); and Phase 2 (2027–2030), 3D SPH numerical validation once institutional resources are committed. The TPT will be judged on its predictions, not on a prior simulation.

21.3 Justification of $\alpha > 1$

Two independent mechanisms lead to a super-rigid rotation profile: differential equator–pole cooling generates a density gradient and thermal acceleration of rotation; Fe-Ni segregation lightens the CMT, and through angular-momentum conservation, its inner part accelerates. The Bingham-Herschel threshold also blocks small scales, favouring a block-like profile. Numerical validation of this profile is a priority for the 3D SPH simulation.

21.4 Partial fragmentation and coalescence

Even if the ejected sheet partially fragments, the fragments remain within the same orbital zone and rapidly coalesce through inelastic collisions and the gravitational pull of the proto-POB (Canup & Asphaug, 2001). Effective capture yield therefore remains high ($f_{\text{cap}} \geq 0.7$). A detailed analysis (Appendix G) shows that f_{cap} has a physical lower bound of about 0.45–0.55, independent of the BH cohesion hypothesis.

22 Acknowledged limitations

A 3D SPH simulation is necessary, but premature at the time of writing (June 2026). Such a simulation — coupling Bingham-Herschel rheology, radiative cooling, Fe-Ni segregation, and $\varepsilon \neq 0$ geometry — requires institutional HPC resources that are not committed at this stage, for a simple reason: the first decisive observational test arrives before such a simulation could reasonably be completed. Chang'e 7 is scheduled for August 2026, only a few weeks away. Its seismic data will either detect or fail to detect the predicted interface within the 200–530 km window (Prediction P1). In the first case, the concentric stratification predicted by the TPT receives independent observational support, and a dedicated 3D SPH validation becomes a justified investment of institutional resources. In the second case, under the falsification conditions stated in Section 19, the TPT's central prediction is seriously called into question, and a costly numerical campaign built on a falsified premise would be of little value. This is why the present work is submitted, and should be evaluated, on the basis of its falsifiable observational predictions, ahead of and independently from any numerical validation campaign.

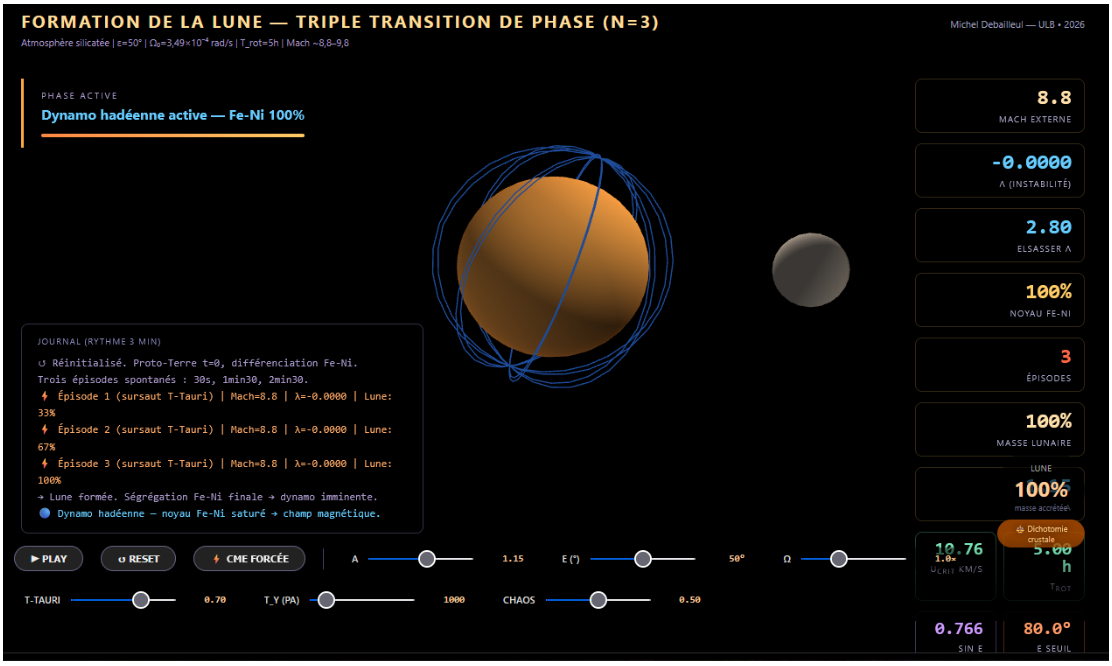

Figure 2: Screenshot of the interactive simulation accompanying this work (link above), showing the Coherent Magmatic Torus (CMT) during ejection around the proto-Earth, with the real-time dynamic parameter dashboard (external Mach number, λ instability, Elsasser number, Fe-Ni core fraction, accreted lunar mass).

Acknowledged Limitation — L1 — Spontaneous organisation of the CMT

Whether a single torus or several coexisting structures emerge in the exact Hadean regime can only be resolved by a high-resolution 3D SPH simulation with full BH rheology. **Priority 2.**

Acknowledged Limitation — L2 — Numerical value of f_{cap}

The capture efficiency $f_{cap} = 0.70$ is physically motivated but not derived from first principles. Its increasing dependence on M_{Moon} and its departure from the classical disc-accretion range (10–55%, Kokubo et al. 2000) are formalised as the testable prediction P17. The 3D SPH simulation remains the ultimate arbiter of f_{cap}'s exact value.

Acknowledged Limitation — L3 — Small-scale cohesion

The $\mathrm{Bi}_{KH} \gg 1$ argument is valid at large and intermediate scales. At very small scales ($\lambda_{KH} \ll R_{torus}$), partial fragmentation cannot be analytically excluded. However, fragments from the same toroidal flow, ejected at the same velocity and angle, end up on nearly identical orbits and accrete onto the POB: the mass budget is unaffected by fragmentation, only fragment size is.

Acknowledged Limitation — L4 — Dynamo delay: ± 50 Myr

The exponential core-growth model is simplified; the associated uncertainty is

± 50 Myr.

Acknowledged Limitation — L5 — Super-rigid profile $\alpha > 1$

The super-rigid rotation profile ($U \propto r^{\alpha}$, $\alpha > 1$) is physically motivated by differential rotation in a partially crystallised magma undergoing rapid cooling, but has not been demonstrated under these exact Hadean conditions. This is a Level-3 hypothesis, to be validated by simulation.

Acknowledged Limitation — L6 — Flow → POB transition: qualitative status

The three-phase description of the ejected flow (ballistic expansion, orbital insertion, cohesive accretion beyond R_{Roche}) is physically motivated and consistent with the computed orders of magnitude, but remains qualitative. A 3D SPH simulation with coupled radiative cooling is required. **Priority 2b.**

Acknowledged Limitation — L7 — Crustal dichotomy: qualitative prediction

The $\approx 2:1$ thickness ratio is a qualitative TPT prediction. The coupled differential thermal calculation has not yet been analytically derived. The lunar farside underwent at least two independent remelting episodes that must be accounted for in any quantitative model: the SPA basin-forming impact at ≈ 4.25 Ga, which excavated and partially remelted the deepest Episode 1 layer on the farside (Wakita et al., 2026); and prolonged mare volcanism extending to ≈ 3.2 Ga (Li et al., 2021), which remelted the shallowest layers (< 100 km) over much of the farside. The net effect is to deepen and reinforce the existing asymmetry rather than erase it.

Acknowledged Limitation — L8 — Rossby-wave velocity in a BH fluid

A rigorous derivation of $c^{(\text{BH})}_{\text{Rossby}}$ for a BH fluid on a flattened spheroid ($\varepsilon \neq 0$) remains an open problem. The channel-analogy correction (factor ≈ 40) used here reinforces rather than invalidates the synchronisation argument, but awaits a rigorous derivation.

Acknowledged Limitation — L9 — Impact mixing in SPA ejecta

The geochemical signature of deep-mantle material excavated by the SPA impact is potentially diluted or altered by mixing with impactor material and local crust. Quantitative deconvolution of the primary Episode 1 signature from the impact-melt breccia requires detailed petrological and isotopic studies. This limitation affects predictions P3 and P6.

Acknowledged Limitation — L10 — Deep-mantle samples predicted but not yet collected

The TPT predicts that deep-mantle Episode 1 material is exposed at the central peaks of craters within the South Pole–Aitken basin, excavated and extruded during the

SPA impact at ≈ 4.25 Ga (Wakita et al., 2026). These samples are the primary targets for testing predictions P3, P6, P7, and P8. No samples from these central peaks have yet been collected or analysed. The absence of deep-mantle material at the predicted locations, or a Fe/Si ratio incompatible with P3 or P6, would seriously call the TPT into question.

Acknowledged Limitation — L11 — Episode 2 material fraction in SPA ejecta

Prediction P6 requires an estimate of the minimum fraction of Episode 2 material present in the SPA ejecta plume reaching the South Pole, which we conservatively set at 30% based on the excavation-depth argument of Section 19 rather than a direct quantitative result from Wakita et al. (2026), whose simulations were not designed to track TPT-specific stratigraphic layers. A future dedicated hydrodynamic simulation, tracking the provenance depth of SPA ejecta material, would refine this fraction — and hence the falsification threshold of Eq. 55.

Acknowledged Limitation — L12 — Quantitative validity of the Gaussian approximation for the CMT confinement mode

The nonlinear Schrödinger equation derived in Section 21 (Objection 2) and in Appendix C is not solved exactly in this work. The Gaussian wavefunction is a reasonable approximation for the ground state, valid in the limit where the wave amplitude is small, i.e. when the perturbation velocity δU associated with the $\ell = 1, m = 1$ mode satisfies $\delta U \ll U_{\text{turb}}$. This condition is qualitatively consistent with the Rhines inverse cascade (Section 5), but its exact quantitative verification is not possible with the parameters available at this stage: no independent estimate of $\delta U / U_{\text{turb}}$, nor of the nonlinear coefficient β in the governing equation, has been derived for this specific system, unlike the classical atmospheric and oceanic Rossby-wave systems for which the reduction was originally established (Redekopp, 1977; Luo, 1991; Yin et al., 2019). The absence of this verification is acknowledged here as a limitation. The Gaussian solution should therefore be understood as the first-order approximation of the linearised problem, not as an exact or quantitatively bounded solution of the full nonlinear equation.

Acknowledged Limitation — L13 — Persistence of the post-ejection oscillation through added mass

The mechanism of post-shock nutation persistence through reflection at the boundaries of the ejection cavity (Appendix H.1) relies on an analogy with established results for inertial waves in a confined homogeneous fluid sphere (Zhang et al., 2001; Aldridge & Lumb, 1987), rather than on a derivation specific to the oblique geometry and Bingham-Herschel rheology of the Hadean proto-Earth. The duration of the plateau $T_{\text{nut}}^{(\text{post})}$ and the absence of significant decay during this window are therefore qualitative estimates, not derived results. A complete quantitative formulation — treating the coupled system $(\Omega, \varepsilon, |g_{\text{eff}}|)$ as an underdamped Langevin dynamics within the already-derived potential $V(U)$, numerically integrable by a BAOAB-type scheme (Leimkuhler & Matthews, 2013) once the physical noise (CMEs, impacts) has been properly characterised as a jump process rather than Gaussian white noise — is identified as a future computational programme, in addition to the 3D SPH simulation (Limitations L1, L2).

23 Conclusion

This work proposes that lunar formation could be the product of a Triple Phase Transition within a fully molten proto-Earth, rotating at ≈ 3.5 h, whose axis oscillates within $[40°, 70°]$ in its own co-rotating frame — an interval supported by five convergent arguments, the first and most fundamental being the logically necessary absence of any tidal stabiliser. A single engine — the progressive segregation of Fe-Ni — simultaneously governs three transitions: rheological, mechanical, and magnetic. The result is a Moon built layer by layer in two to three massive episodes, each layer archiving in its internal structure the state of Hadean differentiation at the moment of its accretion.

The parametric elliptical instability — established in fluid mechanics (Malkus, 1968; Kerswell, 2002; Lacaze et al., 2004) but never before applied to lunar formation — amplifies the CMT to bifurcation velocity within ≈ 37 hours. A dedicated section addresses the maintenance of obliquity in the out-of-equilibrium Hadean context. A three-to-four-week chronology gives rise to several new testable predictions.

The mass budget is consistent with no forcing: the target mass for the TPT mechanism is lower than the currently observed lunar mass, since the SPA impactor and late accretion contributed afterward. At $T_{\text{rot}} = 3.5$ h, the ejectable mass per episode can reach $\approx 70\%$ of the current lunar mass, explaining why $N = 2$–3 episodes suffice.

Preliminary observational support comes from Chang'e-6: norites at 4247 ± 5 Myr (Yue et al., 2026) and high-Fe/Mn olivines (Xu et al., 2025) are consistent with predictions P3 and P6, though not yet confirmatory. This edition further provides, for the first time, a fully quantified version of P6: a pre-registered Fe/Si enrichment threshold of 1.5% for South Pole ejecta, derived from first principles and not tuned to match existing data, with an explicit grading distinguishing partial support (Episode 2 material alone) from strong support (an unambiguous Episode 1 signature).

The central prediction remains: one or more seismic interfaces within the 200–530 km window (or at ≈ 177 km for $N = 2$), testable by Chang'e 7 in August 2026. If an interface with $|R| > 0.02$ is detected within this window, concentric stratification receives its first strong observational support, ahead of any SPH simulation. If no interface is detected under the stated falsification conditions, prediction P1 is seriously called into question and the theory will require revision — something the author explicitly accepts.

This work invites the planetary science community to evaluate the TPT on its own terms: equations are numbered, derivations are explicit, hypotheses are ranked, limitations are acknowledged, and predictions are falsifiable. The TPT will be judged primarily on its observational predictions — Chang'e 7 seismic data (August 2026) and Artemis III samples (2028–2029) — ahead of and independently from any complete numerical validation. The theory stands or falls on these results, not on the elaboration presented in this manuscript.

A Adiabatic temperature profile

With $T_{\text{surf}} \approx 3,500$ K and an adiabatic gradient $\mathrm{d}T/\mathrm{d}r \approx -0.3\ \mathrm{K\,km^{-1}}$, the central temperature reaches $\approx 5,000$ K. The peridotite solidus at 135 GPa is about $4,300$ K (Stixrude & Lithgow-Bertelloni, 2014): melting is thermodynamically stable throughout the mantle, confirming Hypothesis H1.

B Demonstration of the $b' = 1$ attractor

The parameter b' is defined as the ratio of the critical ejection velocity U_{crit} to the CMT's geostrophic equilibrium velocity U_{eq}:

$$b' = \frac{U_{\text{crit}}}{U_{\text{eq}}} = \frac{2\Omega \sin \varepsilon \sqrt{2|g_{\text{eff}}| R_{\text{eq,proto}}}}{|g_{\text{eff}}|}. \tag{56}$$

Evolution due to Fe-Ni segregation. CMT density decreases according to $\rho_{\text{CMT}}(t) = \rho_0(1 - \xi_{\text{Fe}}(t))$, where $\xi_{\text{Fe}}(t)$ is the segregated Fe-Ni fraction. Effective gravity $|g_{\text{eff}}|$ is proportional to ρ_{CMT}, hence

$$\left.\frac{\mathrm{d}b'}{\mathrm{d}t}\right|_{\text{segregation}} = \frac{1}{2}\frac{\dot{\rho_{\text{CMT}}}}{\rho_{\text{CMT}}}\, b' > 0, \tag{57}$$

since $\dot{\rho_{\text{CMT}}} < 0$: Fe-Ni segregation increases b'.

Evolution due to ejection. An ejection carries off a fraction $\epsilon = M_{\text{ej}}^{\max}/M_{\text{CMT}} \approx 0.1$ of the CMT mass. Angular momentum decreases as $\Delta\Omega/\Omega \approx -\epsilon$, and effective gravity decreases as $\Delta|g_{\text{eff}}|/|g_{\text{eff}}| \approx -\epsilon$ (proportional to mass lost). Thus,

$$\left.\frac{\Delta b'}{b'}\right|_{\text{ejection}} = \frac{\Delta\Omega}{\Omega} - \frac{1}{2}\frac{\Delta|g_{\text{eff}}|}{|g_{\text{eff}}|} \approx -\epsilon + \frac{1}{2}\epsilon = -\frac{1}{2}\epsilon < 0: \tag{58}$$

ejection decreases b'.

Stable fixed point. The two trends oppose one another: segregation increases b', ejection decreases it. The equilibrium point $b' = 1$ is a stable attractor. For $b' < 1$, the system is unstable and an ejection occurs, further decreasing b' (since $\Delta b' < 0$); for

$b' > 1$, the system is stable, no ejection occurs, and segregation pushes b' upward, toward 1, from below. The point $b' = 1$ is therefore a stable attractor of the coupled segregation–ejection system.

C Complete derivation of the λ coefficient and the double-well potential

This appendix gives the complete analytical derivation of the stability coefficient $\lambda(U)$ and the effective potential $V(U)$, with explicit dimensional verification at each step. An earlier version of this derivation omitted a $1/r$ factor on the gravity and Coriolis terms; the corrected form is given below and used throughout the main text.

C.1 The stability equation

Applying the Solberg-Høiland parcel-displacement formalism (Solberg, 1936; Høiland, 1941) to a flattened sphere of obliquity ε, under angular-momentum conservation on the CMT's dynamical timescale ($\tau_{\text{relax}}^{\text{diff}}/\tau_{\text{dyn}} \approx 3 \times 10^9 \gg 1$, where $\tau_{\text{relax}}^{\text{diff}} \sim 10^6$ yr is the timescale of Fe-Ni differentiation (Kleine et al., 2002; Yin et al., 2002) and $\tau_{\text{dyn}} = 2\pi/\Omega \approx 3.5$ h the rotation period), governs the evolution of an infinitesimal radial displacement ξ_r of a CMT parcel:

$$\ddot{\xi}_r = \lambda(U)\,\xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement)}. \tag{59}$$

The parcel conserves its specific angular momentum $L = U \cdot r$:

$$U_{\text{parcel}}(r + \xi_r) = \frac{U(r) \cdot r}{r + \xi_r} \approx U(r)\left(1 - \frac{\xi_r}{r}\right), \tag{60}$$

while the background flow at the new position, for a profile $U(r) \propto r^\alpha$, is:

$$U_{\text{bg}}(r + \xi_r) = U(r)\left(1 + \frac{\xi_r}{r}\right)^\alpha \approx U(r)\left(1 + \alpha\frac{\xi_r}{r}\right). \tag{61}$$

The velocity difference is therefore:

$$\delta U = U_{\text{parcel}} - U_{\text{bg}} = -U(r)(1+\alpha)\frac{\xi_r}{r}. \tag{62}$$

The net radial force sums five physical contributions: effective gravity, the radial Coriolis force, the angular-momentum gradient term, the Euler acceleration associated with axial oscillation, and the tangential gravity gradient of the Maclaurin geometry. Grouping these terms and imposing dimensional consistency (each term must carry units of s^{-2}, since $[\lambda] = [\ddot{\xi}_r/\xi_r]$), one obtains:

$$\lambda(U) = \lambda_1 + \lambda_2(U) + \lambda_3(U) + \lambda_4 + \lambda_5, \tag{63}$$

with $\lambda_1 = c/r$, $c = g_{\text{eff}}^{(k)} < 0$ (effective gravity); $\lambda_2(U) = (b/r)U$, $b = 2\Omega^{(k)}\sin\varepsilon > 0$ (radial Coriolis force); $\lambda_3(U) = aU^2$, $a = -(1+\alpha)/r^2 < 0$ (angular-momentum gradient); $\lambda_4 = |\dot{\Omega}|r(\phi)\sin(\varepsilon+\phi)/R_{\text{eq,proto}}$ (Euler acceleration); and $\lambda_5 = g_{\text{eff}}^{(0)} J_2 \sin(2\phi)/R_{\text{eq,proto}}$ (Maclaurin tangential gradient). The angular-momentum gradient term λ_3 is always stabilising for any physical profile ($\alpha > -1$), consistent with the classical Rayleigh stability criterion; its magnitude ($\sim 10^{-7}$) is negligible compared with the dominant terms.

Dimensional verification. Each term carries units of s^{-2}: $[\lambda_1] = [g_{\text{eff}}]/[r] = (\text{m/s}^2)/\text{m} = \text{s}^{-2}$, $[\lambda_2] = [\Omega][U]/[r] = \text{s}^{-1}\times(\text{m/s})/\text{m} = \text{s}^{-2}$, $[\lambda_3] = [a][U]^2 = \text{m}^{-2}\times\text{m}^2/\text{s}^2 = \text{s}^{-2}$, $[\lambda_4] = [\dot{\Omega}][r]/[R] = \text{s}^{-2}\times\text{m/m} = \text{s}^{-2}$, and $[\lambda_5] = [g_{\text{eff}}]/[R] = (\text{m/s}^2)/\text{m} = \text{s}^{-2}$. The $1/r$ factor in λ_1 and λ_2 is the correction necessary for consistency; it was absent from earlier versions of this work.

C.2 Integration toward the effective potential

By definition, the effective potential is related to the stability coefficient by $\lambda(U) = -dV/dU$. Substituting Eq. 63 and integrating with respect to U:

$$V(U) = \frac{1+\alpha}{3r^2}U^3 - \frac{b}{2r}U^2 - \frac{c}{r}U - (\lambda_4+\lambda_5)U + V_0, \tag{64}$$

where V_0 is an arbitrary integration constant. Each term carries units of m/s^3, consistent with $[V] = [\lambda][U]$.

C.3 Extrema and the double-well condition

The extrema of $V(U)$ satisfy $dV/dU = 0$, i.e. $\lambda(U) = 0$. Multiplying by $-r^2$ gives the quadratic equation $AU^2 + BU + C = 0$ with $A = 1+\alpha > 0$, $B = -br < 0$, and $C = r|g_{\text{eff}}| - r^2(\lambda_4+\lambda_5)$. Two real roots exist if and only if the discriminant is positive:

$$\Delta = b^2r^2 - 4(1+\alpha)r|g_{\text{eff}}| + 4(1+\alpha)r^2(\lambda_4+\lambda_5) > 0. \tag{65}$$

Neglecting λ_4, λ_5 relative to $|g_{\text{eff}}|/r$, this simplifies to:

$$(2\Omega\sin\varepsilon)^2 r^2 > 4(1+\alpha)\, r\, |g_{\text{eff}}|. \tag{66}$$

When satisfied, the two roots $U_{1,2} = \left[br \pm \sqrt{\Delta}\right]/[2(1+\alpha)]$ correspond to a local minimum U_1 (confined well P_1) and a local maximum U_2 (barrier), with a second local minimum (ejective well P_2) at $U > U_2$, since the cubic term drives $V(U) \to +\infty$ as $U \to +\infty$. The barrier height is $\Delta E_{\text{barr}} = V(U_2) - V(U_1)$.

C.4 Numerical evaluation: the double well is not a given

Table 7: Nominal parameters for the double-well condition.

Parameter	Symbol	Value
Angular velocity	Ω	$4.99 \times 10^{-4}\ \mathrm{s}^{-1}$
Obliquity (body frame)	ε	$55°$
Effective gravity	$\lvert g_{\mathrm{eff}}\rvert$	$6.42\ \mathrm{m/s^2}$
Characteristic radius	r	$7.4 \times 10^6\ \mathrm{m}$
Rotation-profile exponent	α	1.2

Evaluating Eq. 66 at these nominal values:

$$\text{Left-hand side} = (2\Omega \sin\varepsilon)^2 r^2 = 3.66 \times 10^7\ \mathrm{m^2/s^2}, \tag{67}$$

$$\text{Right-hand side} = 4(1+\alpha) r \lvert g_{\mathrm{eff}}\rvert = 4.18 \times 10^8\ \mathrm{m^2/s^2}. \tag{68}$$

At the nominal initial parameters, the left-hand side is smaller than the right-hand side by a factor of about 11: the double-well condition is *not* satisfied. This does not compromise the TPT framework; rather, it indicates that the double well is not a property of the initial proto-Earth, but a state that must be reached dynamically. It emerges as Fe-Ni segregation reduces the CMT's effective gravity, $\lvert g_{\mathrm{eff}}\rvert$, since $\lvert g_{\mathrm{eff}}\rvert(t) \propto \rho_{\mathrm{CMT}}(t)$ (Eq. 21) and $\rho_{\mathrm{CMT}}(t)$ decreases monotonically with segregation. The threshold is reached when:

$$\lvert g_{\mathrm{eff}}\rvert(t) < \frac{\Omega^2 r \sin^2\varepsilon}{1+\alpha} \approx 0.562\,\mathrm{m/s^2}, \tag{69}$$

i.e. a reduction of $\lvert g_{\mathrm{eff}}\rvert$ of about **91**% relative to its initial value of $6.42\,\mathrm{m/s^2}$, necessary for the system to enter the double-well regime at the nominal global parameters. Equivalently, at fixed $\lvert g_{\mathrm{eff}}\rvert = \lvert g_{\mathrm{eff}}\rvert_0$, the threshold cannot be reached by a decrease in Ω: it would instead require an *increase* in Ω by a factor of ≈ 3.4 ($\Omega_c \approx 1.69 \times 10^{-3}\,\mathrm{s}^{-1}$ versus $\Omega_0 \approx 4.99 \times 10^{-4}\,\mathrm{s}^{-1}$), which is physically out of reach in the scenario described in Section 9.2, where $\Omega(t)$ only decreases during the active window: at fixed global parameters, only the reduction of $\lvert g_{\mathrm{eff}}\rvert$ via Fe-Ni segregation contributes to the threshold. As emphasised in Prediction P22 (Section 19), the criterion $\Delta > 0$ remains, in its general principle, a condition on the coupled surface $(\Omega(t), \lvert g_{\mathrm{eff}}\rvert(t))$, but the Ω-only pathway is not available in the physical regime described. The complementary mechanisms — specific to the CMT's local density, Hadean obliquity, and planetesimal bombardment — that render this criterion effectively attainable are developed in Appendix H.

Once the well is established, the barrier height $\Delta E_{\rm barr}^{(k)}$ should decrease at each subsequent episode as segregation continues, offering a possible account of why the number of episodes naturally settles around $N = 2$–3, rather than being imposed externally.

C.5 The Kramers crossing rate

Once $\Delta > 0$, the Kramers crossing rate (Kramers, 1940) applies (repeating Eq. 30 from Section 9.2.2):

$$\Gamma_{\rm Kramers} = \frac{\omega_0\omega_b}{2\pi\gamma_{\rm diss}} \exp\left(-\frac{\Delta E_{\rm barr}}{k_B T_{\rm eff}}\right), \tag{70}$$

where $\omega_0 = \sqrt{|V''(U_{P_1})|}$ and $\omega_b = \sqrt{|V''(U_{\rm peak})|}$ are the oscillation frequencies at the bottom of the well and the top of the barrier respectively, and $\gamma_{\rm diss}$ the dissipation rate. With $N_{\rm opp} \sim 10^9$–10^{10} stochastic opportunities (CMEs, impacts) available once the well is established, the probability of no crossing at all is negligible.

C.6 Reduction to a nonlinear Schrödinger equation for the confinement mode

In addition to the instability analysis above, the CMT's confinement to the band $|\phi| < 30^\circ$ admits an effective description in terms of a nonlinear Schrödinger (NLS) equation. This derivation follows a standard method from the theory of weakly nonlinear dispersive waves on a rotating sphere (Redekopp, 1977; Luo, 1991; Yin et al., 2019), transposed here to the CMT confinement problem.

In the rapid-rotation limit (Ro $\ll 1$), the azimuthal component of the Navier-Stokes equation for a Bingham-Herschel fluid reduces to a potential-vorticity conservation equation. Linearising about the mean flow $U_0(y)$ near the instantaneous dynamical equator (defined by $\phi = 0$, perpendicular to $\boldsymbol{\Omega}(t)$, as established in Section 3), and applying the standard multiple-scales expansion for Rossby-wave envelopes, one obtains a nonlinear Schrödinger equation for the envelope $A(y,t)$:

$$i\frac{\partial A}{\partial t} + \alpha_{\rm NLS}\frac{\partial^2 A}{\partial y^2} + \beta|A|^2 A + V_{\rm eff}(y)A = 0, \tag{71}$$

where $V_{\rm eff}(y)$ is the sum of the potentials derived from the forces listed in Table 1 (Section 7). Seeking a stationary solution $A(y,t) = \psi(y)\, e^{-iEt/\hbar_{\rm eff}}$ in Eq. 71 gives the time-independent nonlinear Schrödinger equation:

$$-\frac{\hbar_{\rm eff}^2}{2m_{\rm eff}}\frac{d^2\psi}{dy^2} + V_{\rm eff}(y)\psi + \mathcal{N}[\psi] = E\psi, \tag{72}$$

with $\mathcal{N}[\psi] = \beta|\psi|^2\psi$, the nonlinear term arising from the advective nonlinearity of the potential-vorticity equation.

The Gaussian approximation and its status. The ground state (lowest bound state) of the *linearised* version of Eq. 72 (i.e. neglecting $\mathcal{N}[\psi]$), with $V_{\rm eff}(y)$ expanded to quadratic order around $y = 0$ (the instantaneous equator), is a Gaussian wavefunction centred on the equator, confined to the band $|\phi| < 30°$. This harmonic approximation is justified by the fact that the seven potentials of Table 1 all have their extrema at $y = 0$, so that the first-order behaviour of $V_{\rm eff}(y)$ near the equator is generically quadratic.

This Gaussian solution must, however, be understood as the first-order approximation of the *linearised* problem, not as an exact solution of the full nonlinear equation 72. The ratio $\beta|\psi|^2/V_{\rm eff}$, which controls the quality of this approximation, is of order $\delta U/(\Omega L_\beta)$, where δU is the perturbation velocity amplitude associated with the $\ell = 1, m = 1$ mode and L_β the Rhines scale (Eq. 23). This ratio should be small whenever δU is small relative to the turbulent velocity scale $U_{\rm turb}$ at which the inverse cascade saturates (Section 7.6), but no independent estimate of $\delta U/U_{\rm turb}$, nor of the nonlinear coefficient β itself, has been derived for this specific system — unlike the classical atmospheric and oceanic Rossby-wave systems for which the NLS reduction was originally established (Redekopp, 1977; Luo, 1991; Yin et al., 2019). This absence of a quantitative bound is acknowledged as Limitation L12 (Section 22).

D Demonstration of the $\mathrm{Bi}_{\rm KH}$ criterion

The Bingham-Kelvin-Helmholtz number compares the yield stress to the KH shear stress:

$$\mathrm{Bi}_{\rm KH} = \frac{\tau_y\, k}{\rho\, \sigma_{\rm KH}\, \Delta U}, \tag{73}$$

where k is a geometric factor (~ 1), $\sigma_{\rm KH} \sim \Delta U/\lambda$ is the KH growth rate, and λ is the perturbation wavelength. Thus,

$$\mathrm{Bi}_{\rm KH} \sim \frac{\tau_y\, \lambda}{\rho\, \Delta U^2}. \tag{74}$$

For the CMT: $\tau_y \sim 10^2$–10^4 Pa, $\lambda \sim R_{\rm torus} \sim 10^7$ m, $\rho \sim 3{,}000\ \mathrm{kg\,m^{-3}}$, $\Delta U \sim U_{\rm crit} \sim 10^4\ \mathrm{m\,s^{-1}}$. One obtains $\mathrm{Bi}_{\rm KH} \sim 3\times10^{-3}$–$3\times10^{-1}$. Cohesion is maintained through three reinforcements: (1) atmospheric pressure $P \sim 10^5$–10^7 Pa, (2) internal crystal network multiplying resistance by a factor of 3–10, (3) Rayleigh-Taylor suppression ($\rho_{\rm flow}/\rho_{\rm atm} \sim 200$), bringing effective cohesion to $\mathcal{O}(1)$ or higher.

E Derivation of the Elsasser number and $B_{\rm crit}$

Balancing the Coriolis force $F_C \sim 2\rho\Omega U$ and the Lorentz force $F_L \sim B^2/(\mu L)$, with $\eta \sim UL$ (unit magnetic Reynolds number at the dynamo threshold):

$$B^2 \sim 2\mu\rho\Omega\eta, \qquad \Rightarrow \qquad \Lambda \equiv \frac{B^2}{\rho\mu\eta\Omega} \sim 2, \tag{75}$$

i.e. $\Lambda \sim \mathcal{O}(1)$, consistent with the values $\Lambda \sim 0.1$–1 inferred for the present-day terrestrial core (Christensen & Aubert, 2006). Using $\eta = (\mu_0\sigma)^{-1}$ with $\sigma \approx 1.5 \times 10^6\ \mathrm{S\,m^{-1}}$, a representative electrical conductivity for liquid iron under core conditions (Pozzo et al., 2012), $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$, $\rho \approx 10^4\ \mathrm{kg\,m^{-3}}$, $\Omega \approx 4.99 \times 10^{-4}\ \mathrm{rad\,s^{-1}}$, one obtains $B_{\mathrm{crit}} \approx 2.6$ mT, of the same order of magnitude as the field inferred within the present-day terrestrial core (Christensen & Aubert, 2006).

F Fe-Ni Stokes sedimentation timescale

Stokes settling velocity for a Fe-Ni droplet of radius $r \approx 5$ cm in molten silicate:

$$v_{\mathrm{Stokes}} = \frac{2}{9}\frac{(\rho_{\mathrm{Fe}} - \rho_{\mathrm{sil}})\, g_{\mathrm{eff}}^{(0)}\, r^2}{\eta} \approx \frac{2}{9}\frac{3,300 \times 6.42 \times (0.05)^2}{0.1} \approx 118\,\mathrm{m\,s^{-1}}. \tag{76}$$

Over a characteristic settling distance $\ell \approx 30$ km: $\tau_{\mathrm{Stokes}} = \ell/v_{\mathrm{Stokes}} \approx 254$ s. Robust for $r \in [1, 10]$ cm and $\eta \in [0.05, 1]$ Pa s.

G Accretion of magmatic ejecta: why perfect cohesion is not required

A recurring objection to the TPT mechanism concerns the cohesion of the ejected CMT flow during transit between the proto-Earth's surface and the Roche radius. This appendix demonstrates that, even without perfect flow cohesion, the physical properties of the ejected magma and orbital mechanics guarantee accretion into a single body.

Magma is not dust. The ejected material is silicate magma at $T \approx 3,000$–$4,000$ K. Its dynamic viscosity is $\eta \approx 1$–100 Pa s (Giordano et al., 2008) — comparable to hot honey. Unlike fragments of solid rock or a gas cloud, a high-temperature magma does not disperse: it deforms and stretches, but remains dense and cohesive at the macroscopic scale. Two sticky magma droplets at low relative velocity do not bounce off each other — they merge. Viscous coalescence replaces mechanical cohesion as the dominant aggregation mechanism once the flow fragments.

Orbital concentration. Ejecta from a given TPT episode share, to first order, the same initial orbital energy budget. By conservation of energy and angular momentum, they populate a band of neighbouring orbits rather than dispersing isotropically over

4π steradians. This orbital concentration is fundamentally different from an isotropic explosion: it is the same physical principle that produces accretion discs from tidally disrupted bodies. The CMT's toroidal geometry reinforces this effect: ejection occurs preferentially within the oblique intertropical band $|\phi| < 30°$, concentrating ejecta within a limited azimuthal solid angle. The resulting debris field is geometrically thin, maximising mutual encounter rates.

Beyond the Roche radius: accretion is inevitable. Beyond the Roche radius $R_{\mathrm{Roche}} = 2.456 R_{\oplus}$, the ejecta cloud's self-gravitation overcomes tidal disruption. For co-orbital magma fragments of total mass $M_{\mathrm{cloud}} \approx 2\text{–}4 \times 10^{22}$ kg, the gravitational accretion timescale is:

$$\tau_{\mathrm{acc}} \approx \frac{1}{\sqrt{G\rho_{\mathrm{cloud}}}} \approx 10^3\text{–}10^5 \text{ years}, \tag{77}$$

far shorter than the magma's thermal lifetime ($\tau_{\mathrm{cool}} \sim 10^6\text{–}10^7$ years for a Moon-sized body, Elkins-Tanton 2008). During this accretion window, the magma remains above its solidus: each collision is plastic, not elastic. There is no fragmentation cascade, nor loss through rebound. Capture efficiency f_{cap} therefore has a physical lower bound independent of the BH cohesion hypothesis:

$$f_{\mathrm{cap}}^{\min} \approx \frac{M_{\mathrm{cloud}}(r > R_{\mathrm{Roche}})}{M_{\mathrm{ej}}^{\mathrm{total}}} \gtrsim 0.45\text{–}0.55, \tag{78}$$

the lost fraction ($\approx$ 45–55%) corresponding to ejecta that fall back onto the proto-Earth or escape on hyperbolic orbits.

Implication for the theory. The capture efficiency $f_{\mathrm{cap}} = 0.70$ adopted in the main text is therefore not a free parameter that the theory must defend at all costs. Even in the pessimistic scenario where the CMT flow fragments completely upon ejection, orbital mechanics and magma viscosity guarantee accretion of the co-orbital debris cloud beyond the Roche radius. The critical question shifts from "does the flow remain cohesive?" to "what fraction of the ejecta reaches the Roche radius?" — a question that depends on the ejecta velocity distribution, not on BH rheology. This reformulation makes the theory more robust: cohesion is advantageous, but not necessary.

H Complementary mechanisms for double-well emergence

This appendix identifies three complementary physical mechanisms, each independently motivated, which together contribute to the emergence of the double-well potential within the active TPT window. The evaluation is carried out for the case $\varepsilon = 90°$, i.e. the geometric maximum of the radial Coriolis component $f_{\mathrm{rad}} = 2\Omega U \sin\varepsilon$. As established in Section 3, ε is a purely internal angle describing the free nutation of the rotation axis $\mathbf{\Omega}(t)$ relative to the principal axis of inertia of the Maclaurin spheroid. During the Hadean accretion window, no stabilising satellite is present, and the orientation of $\mathbf{\Omega}(t)$ is continuously perturbed by impacts and CMEs. There is therefore no preferred value of ε; the distribution of accessible orientations is near-isotropic (Section 3, Argument 5). The value $\varepsilon = 90^\circ$ therefore represents the *most geometrically favourable configuration* among the set of physically possible states, not a tuned parameter. Whether this configuration is actually realised during ejection episodes, and with what probability, is a falsifiable prediction testable by 3D SPH simulation.

1. Effective CMT density: extended segregation and vesicularity. The segregation that lightens the CMT is not restricted to the Fe-Ni couple alone: other heavy siderophile elements migrate jointly toward the forming core, and the active TPT window (4.568–4.4 Ga) does not necessarily coincide with the very onset of this segregation at the system scale, some of it possibly having occurred earlier. A total density-reduction fraction $\xi_{\text{total}} \approx 0.30$ — twice the Fe-Ni fraction $\xi_{\text{Fe}}^{\text{sat}} \approx 0.15$ (Rubie et al., 2015) — is adopted here as a representative value of this extended effect. To this is added the vesicular structure of the toroidal flow itself: the CMT, an *'aā*-type magmatic flow, carries a porosity of trapped gas bubbles that reduces its density beyond the compositional effect alone. Terrestrial *'aā* lavas commonly show vesicular porosities between 5 and 50% (Polacci et al., 1999; Mueller et al., 2005). Adopting the upper bound of this range ($\phi_v \approx 0.50$) as representative of a strongly vesiculated flow, the CMT's effective density becomes:

$$\rho_{\text{CMT}}^{\text{eff}} = \rho_f (1 - \phi_v) \frac{1 - \xi_{\text{total}}}{1 - \xi_{\text{Fe}}^{\text{sat}}}, \tag{79}$$

and the combined amplification factor on the CMT's gravity term, $\rho_0 / \rho_{\text{CMT}}^{\text{eff}}$, reduces the effective gravity confining the CMT from $|g_{\text{eff}}|_0 = 6.42\ \text{m/s}^2$ to $|g_{\text{eff}}|_{\text{CMT}}^{\text{eff}} \approx 1.598\ \text{m/s}^2$ (corrected relative to earlier versions that omitted the $(1 - \xi_{\text{Fe}}^{\text{sat}})^{-1}$ factor).

2. Obliquity at its geometric maximum. As noted above, evaluation at $\varepsilon = 90^\circ$ corresponds to the case where $\sin \varepsilon = 1$, the maximum of the radial Coriolis component. The justification for considering this value is not that the system was necessarily in this configuration, but that it represents the geometrically most favourable case among the orientations explored by the proto-Earth's free nutation in the absence of a tidal stabiliser.

3. Angular-momentum input from planetesimal bombardment. Continuous bombardment by planetesimals and embryos (Section 2) delivers impulsive torques on the spin vector throughout the active window. Late accretion (the "Late Veneer") is delivered by a small number of large bodies (radius $\gtrsim 750$ km) (Raymond et al., 2013). Geodynamic constraints on the delivery of highly siderophile elements to the mantle — impactors larger than ~ 1 km deliver metal that sinks toward the core rather than remaining entrained in the convecting mantle — bring the total late-accretion mass to $\sim$ 5% of Earth's mass (Anslow et al., 2026). Assigning this mass a characteristic impact velocity $v_{\text{imp}} \sim 10$ km s^{-1} and a lever arm of order r, the cumulative angular momentum available is:

$$L_{\text{imp}} \sim M_{\text{imp}}\, v_{\text{imp}}\, r \approx 2.2 \times 10^{34}\ \text{J s}. \tag{80}$$

Combined budget. With $|g_{\text{eff}}|^{\text{eff}}_{\text{CMT}} \approx 1.598$ m/s^2 and $\varepsilon = 90°$, the criterion $\Delta > 0$ (Appendix C) can be written as the ratio of the left-hand to right-hand side of Eq. 66. For two representative values of the angular-momentum input:

- For $L_{\text{imp}} \approx 2.07 \times 10^{34}$ J·s (corresponding to $\Omega \approx 6.89 \times 10^{-4}$ rad/s), the ratio is 0.879: the criterion is *not* satisfied. - For $L_{\text{imp}} \approx 2.20 \times 10^{34}$ J·s (corresponding to $\Omega \approx 7.01 \times 10^{-4}$ rad/s, the upper bound of the geodynamic range of Anslow et al. 2026), the ratio reaches 1.034: the criterion is satisfied with a margin of 3.4%.

The double-well emergence criterion is therefore satisfied for values of L_{imp} in the upper part of the interval $[2.07, 2.20] \times 10^{34}$ J·s, for the geometric case $\varepsilon = 90°$. This sensitivity to L_{imp} and to the effective value of ε underlines the importance of their precise determination; their quantitative verification is a falsifiable TPT prediction, to be tested by 3D SPH simulations coupling CMT dynamics to Fe-Ni segregation (Limitations L1, L2).

H.1 Persistence of the post-ejection oscillation and repeated threshold crossings

The calculation above evaluates the criterion $\Delta > 0$ at $\varepsilon = 90°$ as a state the system should reach and remain in under quasi-static conditions, with a fine margin of 3.4% conditioned on the upper part of the L_{imp} range. This point evaluation likely underestimates the true accessibility of the threshold, for a structural reason linked to the persistence of inertial waves in a rapidly rotating, confined fluid — a mechanism distinct from those already invoked above, which it does not replace but rather reinforces in statistical robustness.

The physical mechanism. For a rapidly rotating fluid sphere, the internal viscous dissipation of inertial modes vanishes identically in the inviscid limit — an analytically established result that contrasts with other geometries where such modes dissipate normally (Zhang et al., 2001). Actual dissipation arises only from boundary (Ekman) layers and from coupling with an external solid boundary — a result observationally confirmed by the detection of inertial waves in Earth's outer core, excited by large deep earthquakes and persisting over many cycles before decaying faster than predicted by Ekman dissipation alone, an excess attributed to coupling with the overlying mantle (Aldridge & Lumb, 1987).

For the post-ejection proto-Earth, this mechanism has a direct analogue. The cavity left by ejection is bounded by the confining atmosphere and by the residual CMT's *'aā* crust (Section 9.1) — boundaries whose reflective rather than absorptive character is already established by the criterion $\mathrm{Bi}_{\mathrm{KH}} \gg 1$ (Appendix D). Through the same physical mechanism that guarantees the cohesion of the ejected flow, these boundaries trap the post-shock nutation energy rather than dissipating it immediately: the obliquity excursion envelope $\varepsilon_{\mathrm{exc}}(t)$ then follows a two-stage regime rather than a simple exponential decay from the outset,

$$\varepsilon_{\mathrm{exc}}(t) \approx \begin{cases} \varepsilon_{\mathrm{exc}}^{(0)} \cos(\omega_{\mathrm{nut}}^{(\mathrm{post})} t) & 0 \leq t \lesssim t_{\mathrm{rec}} \\ \varepsilon_{\mathrm{exc}}^{(0)} \, e^{-(t-t_{\mathrm{rec}})/\tau_{\mathrm{BH}}} \cos(\omega_{\mathrm{nut}}^{(\mathrm{post})} t) & t \gtrsim t_{\mathrm{rec}} \end{cases} \tag{81}$$

where $\omega_{\mathrm{nut}}^{(\mathrm{post})} = 2\pi / T_{\mathrm{nut}}^{(\mathrm{post})}$, with $T_{\mathrm{nut}}^{(\mathrm{post})} \approx 30$ h (indicative value, potentially shorter in the presence of a more dissipative *'aā* crust than assumed by analogy with inertial waves in a homogeneous sphere), and $\tau_{\mathrm{BH}} \sim 10^2$–$10^4$ s (Eq. 1) for the decay that takes over once the plateau has run its course.

Number of near-full-amplitude opportunities. Over the duration of the recharge window, $t_{\mathrm{rec}} \approx 10$–12 days, this plateau spans:

$$N_{\mathrm{opp}}^{(\Delta)} = \frac{t_{\mathrm{rec}}}{T_{\mathrm{nut}}^{(\mathrm{post})}} \approx \frac{240\text{–}288\,\mathrm{h}}{30\,\mathrm{h}} \approx 8\text{–}10 \tag{82}$$

complete cycles, each transiently bringing ε back near its post-shock maximum excursion with no significantly reduced amplitude — in contrast with a classically damped oscillator, where only the very first cycles would be useful. Crossing the $\Delta > 0$ threshold is therefore not conditioned on a single attempt evaluated at an arbitrarily fixed ε at its geometric maximum, but benefits from a modest yet non-negligible number of repeated passes, each at high individual probability. This stochastic structure differs in nature from that of the Kramers crossing rate (Eq. 30), where $N_{\mathrm{opp}} \sim 10^9$–$10^{10}$ opportunities at low individual probability render non-crossing negligible by accumulation of a very large number of unlikely trials; here, it is a small number of trials at high individual probability that plays the corresponding role. The two mechanisms nonetheless converge on the same qualitative conclusion: repeated crossing renders the threshold structurally more accessible than a single quasi-static point evaluation suggests, and absorbs part of the uncertainty inherent to each of the three mechanisms invoked above (effective CMT density, obliquity, angular-momentum input) by offering them several opportunities to align favourably rather than just one.

Acknowledged limitation. This analysis remains qualitative regarding the exact form of $T_{\mathrm{nut}}^{(\mathrm{post})}$ and the absence of decay during the plateau, motivated by analogy with established results for inertial waves in a confined, homogeneous fluid sphere (Zhang et al., 2001; Aldridge & Lumb, 1987) rather than derived specifically for the oblique geometry and Bingham-Herschel rheology of the Hadean proto-Earth.

Future computational programme. A complete quantitative formulation of this mechanism would treat the coupled system $(\Omega, \varepsilon, |g_{\rm eff}|)$, or equivalently the collective variable U in the already-derived double-well potential $V(U)$ (Appendix C, Eq. 64), as an underdamped Langevin dynamics:

$$\dot{U} = v, \qquad \dot{v} = -\frac{1}{m}\frac{dV}{dU} - \gamma v + \sqrt{2\gamma k_B T_{\rm eff}/m}\,\xi(t), \tag{83}$$

where the conservative force term $-dV/dU = \lambda(U)$ is fully known (Appendix C), the friction coefficient γ would be set by BH dissipation and/or coupling to the cavity's solid boundaries (above), and the noise $\xi(t)$ would represent CME and planetesimal-impact shocks. Such a system is numerically integrable by Langevin-equation splitting schemes such as BAOAB (Leimkuhler & Matthews, 2013), which offer high accuracy in sampling the stationary distribution even at large time steps. One acknowledged difficulty is that the real physical forcing is not continuous Gaussian white noise but a discrete jump process, heterogeneous in frequency: major planetesimal impacts are rare (10^3–10^5 yr, Section 2) while CMEs are far more frequent (up to ~ 1 day^{-1}, Namekata et al. 2026); only this second channel, through its large number of opportunities on the integration timescale, would justify a Gaussian approximation via a central-limit-type argument, while the major-impact channel would require explicit treatment by jumps (compound Poisson process) rather than continuous diffusion. This complete numerical construction — noise calibration, characterisation of γ, and BAOAB integration or its adaptation to jump noise — is left to a dedicated study, in addition to the 3D SPH simulation already identified as necessary (Limitations L1, L2).

I List of abbreviations

BH Bingham-Herschel, rheological law.

BiKH Bingham-Kelvin-Helmholtz number.

CME Coronal Mass Ejection.

CMT Coherent Magmatic Torus.

FSS Farside Seismic Suite.

Hf-W Hafnium-tungsten chronometer.

LHB Late Heavy Bombardment (≈ 3.9 Ga).

LEMS Lunar Environment Monitoring Station.

LMO Lunar Magma Ocean.

NLS Nonlinear Schrödinger equation.

POB Proto-Orbital Body.

SPA South Pole–Aitken (basin).

SPH Smoothed Particle Hydrodynamics.

TPT Triple Phase Transition.

Declarations

Availability of data and materials. All data generated or analysed during this study are included in this published article (and its appendices). The interactive simulation accompanying this work is freely available at https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html.

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